AIM-ALPHA documentation

Agricultural Land-use Partial Equilibrium Model

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AIM-ALPHA documentation

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Abstract

This document describes the model structure and system of equations for AIM-ALPHA v1.0.5. AIM-ALPHA is a one-year step sequential dynamic partial equilibrium model targeting agricultural markets and land use. It performs future projections considering 166 countries on the demand side, 400 production units on the supply side, 23 agricultural commodities, and major land-use categories such as cropland, pasture, forest, and other natural land. This model is anticipated to be applicable to research including analyses of food security issues, climate change impacts, evaluations of climate change mitigation measures, and linkages to external assessments of biodiversity impacts related to land use.

Keywords: Integrated assessment model, partial equilibrium, agricultural markets, land use, climate change mitigation

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1 Introduction

This document describes the model structure and system of equations of AIM-ALPHA (the Asia-pacific Integrated Model - Agricultural Land-use Partial equilibrium model for Harmonized Analysis), a partial equilibrium model for agriculture and land use. The document is organized as follows:

1) Model Structure (Section 2)
2) System of Equations (Sections 3 and 4)
3) Data Processing (Section 5)
4) Parameter Estimation (Section 6)
5) Model Calculation Flow and Execution Method (Section 7)

This document aims to enhance transparency and reproducibility of the model's analytical components, provide clear instructions for model execution and configuration, facilitate linkage with other models, and support further applications in global and national assessments of sustainable land-use transitions.

2 Model Structure

This section provides an overview of the AIM-ALPHA model. AIM-ALPHA is a recursive-dynamic partial equilibrium model focusing on agricultural markets and land use, operating with an annual time step for the time horizon from 2015 to 2100. The model is formulated as a mixed complementarity problem (MCP) and handles approximately 200,000 variables. The basic model structure is shown in Fig. 1.

In terms of spatial resolution, the demand side covers 166 countries globally, while the production side consists of 400 production units. These production units are defined by overlaying national boundaries [1] with FAO's 230 major river basin boundaries [2]. Figure 2 illustrates the geographical coverage of the countries and production units.

Food production is categorized into 13 primary crops, 6 livestock products, and 5 processed goods. Table 1 lists the food commodity classifications used in the model. The production functions are represented by Leontief-type functions, where each good is produced by combining cropland, feed and pasture, and raw materials as inputs. Other factors such as labor, capital, and energy are aggregated into a single production factor.

Primary crops and livestock products that require land are differentiated by production unit, whereas processed goods, which do not require land inputs, are modeled at the level of the 166 countries. Land use within each production unit is allocated according to a multi-nested logit function that balances total land area, and land shares are determined by relative land prices. The non-pasture land nest includes cropland, forest, other natural land, and afforested land. Protected areas are represented as minimum areas for the corresponding land classes, while the agro-ecological suitability constraint limits cropland expansion.

Food demand is driven by population and GDP, with income elasticity considered for future changes. Food prices are determined endogenously so that supply and demand are balanced in the domestic market. Within each country, the balance between production, domestic demand, and net trade (imports minus exports) is maintained. In the global market, total world imports and exports are balanced after accounting for a calibrated trade residual.

AIM-ALPHA can be linked with other AIM models. AIM-Hub provides socioeconomic, climate-policy, energy-crop, and carbon-price drivers; AFOLUB can provide land-based mitigation assumptions; and AIM-PLUM can return changes in basin-level crop yields resulting from spatial land-use allocation [3, 4, 5].

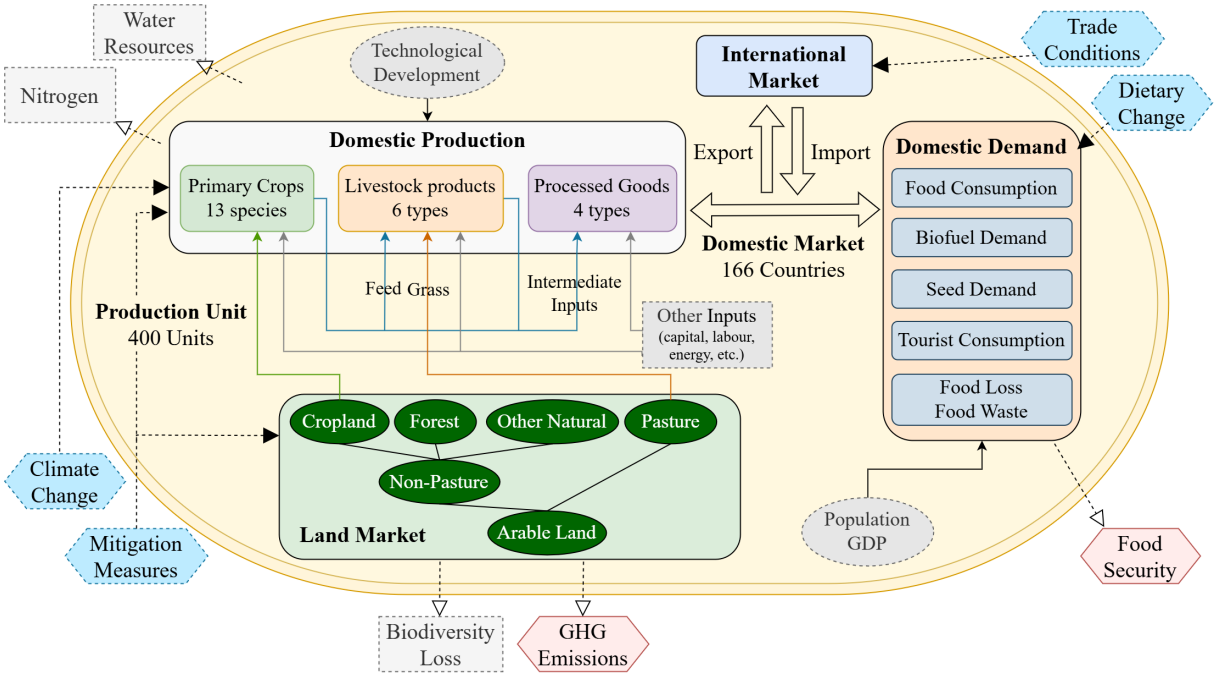


Figure 1: Overview of AIM-ALPHA

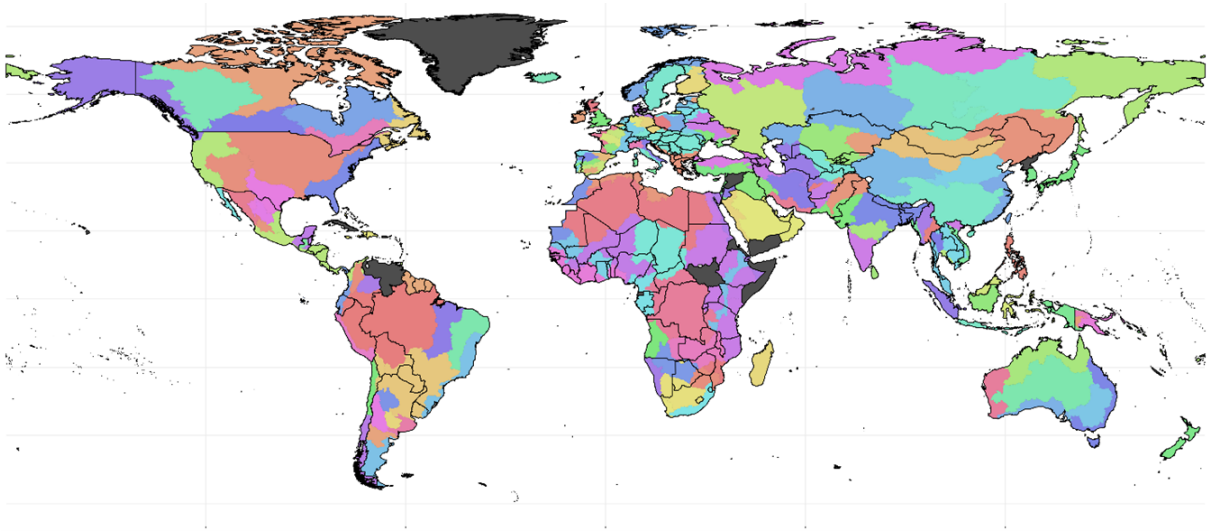


Figure 2: Geographical coverage of countries (166) and production units (400).

Table 1: Classification of food commodities

Category	Commodity	Model code
Primary crops	Wheat	`wht`
	Rice	`rce`
	Maize	`mze`
	Other grains	`crl`
	Starchy roots	`str`
	Sugar crops	`sgr`
	Legumes	`pls`
	Nuts	`nut`
	Oilseeds	`ocr`
	Vegetables	`vgt`
	Fruits	`frt`
	Stimulants	`stm`
	Spices	`spc`
Livestock products	Beef	`cmt`
	Sheep and goat meat	`rmt`
	Poultry meat	`pmt`
	Other meat	`omt`
	Eggs	`egg`
	Raw milk	`mlk`
Processed products	Sugar and sweeteners	`swt`
	Vegetable oils	`vol`
	Oilseed cakes	`cak`
	Alcoholic beverages	`alc`
	Dairy products	`dai`

3 Variables and Parameters

This section describes the variables and parameters used in the model.

3.1 Indices

Table 2 lists the indices used in the model.

Table 2: List of indices

Index	Description	Dimension	Notes
`cty`	Country	166	
`bas`	Production unit	400	Unique country–river-basin unit
`reg`	Macro region	17	Same as the regional classification in AIM-Hub [3]. See Section
`c(a)`	Commodities (production activities)	25	24 food commodities (Table 1) and dedicated energy crops
`a_bas`	Basin-level production activities	20	Crops and livestock
`a_cro`	Primary crops	14	Including dedicated energy crops
`a_liv`	Livestock products	6	
`a_liv_rum`	Ruminant livestock products	3	Cattle meat, other ruminant meat, raw milk
`a_pro`	Processed products	5	
`c_trd`, `c_ntrd`	Traded / non-traded commodities		
`c_fed`	Feed inputs	10	
`c_int`	Intermediate inputs	8	
`c_bio`	Biofuel crops	3	
`wat`	Crop production technology	2	Irrigated / rainfed
`cz`	Climate zone	4	Tropical, temperate, arid, hyper-arid
`sys`	Livestock production system	3	Pasture-based, mixed farming, aggregate non-ruminants
`lu`	Land-use class	7	`cro`, `pas`, `primf`, `secdf`, `gra`, `pastr`, `range`
`lu_prot`	Protected land-use class	5	`primf`, `secdf`, `gra`, `pastr`, `range`

Continued on next page

Table 2: List of indices (continued)

Index	Description	Dimension	Notes
`lu_pas`	Protected pasture class	2	`pastr`, `range`
`src`	GHG emission sources	14	Enteric fermentation, manure management, rice cultivation, synthetic fertilizer application, organic fertilizer application, manure on pasture, cultivation of organic soils, crop residues, crop residue burning, grassland burning, forests, natural wildfires, deforestation, afforestation
`gas`	Gas species	3	CO_2, CH_4, N_2O
`age`	Forest age	100	1 to 100 years

3.2 Endogenous Variables

Table 3 lists the endogenous variables used in the model.

Table 3: List of endogenous variables

Variable	Description	Unit
`QD`	Domestic demand	1000t
`QDFood`	Food consumption demand	1000t
`QDFed`	Feed demand	1000t
`QDInt`	Intermediate input demand for processed goods	1000t
`QDBio`	Biofuel demand	1000t
`QDSeed`	Seed demand	1000t
`QDTour`	Tourist consumption demand	1000t
`QDLoss`	Food loss amount	1000t
`QDWaste`	Food waste amount	1000t
`QDOth`	Other demand	1000t
`QFeed`	Feed input	1000t
`QInt`	Raw material input	1000t
`QS`	Domestic production	1000t
`QSBas`	Production by river basin	1000t
`QSWat`	Crop production by technology	1000t
`QSSys`	Livestock production by system	1000t
`QM`	Imports	1000t
`QE`	Exports	1000t

Continued on next page

Table 3: List of endogenous variables (continued)

Variable	Description	Unit
NT	Net trade	1000t
PP	Producer price	$/t
PC	Consumer price	$/t
PCIdx	Subsidy-adjusted consumer price index	-
PB	Border-equivalent price of the domestic good	$/t
PM	Import price	$/t
PE	Export price	$/t
PW	World price	$/t
QGrass	Grass utilization	1000t
YLiv	Livestock yield per hectare	t/ha
NAnimal	Number of livestock	1000 head
COBas	Unit cost of other production inputs by basin	$/t
COCty	Unit cost of other production inputs by country	$/t
QLVeg	Vegetation area	1000ha
QLPas	Total pasture area	1000ha
QLPasGrz	Grazed pasture area	1000ha
QLPasUng	Ungrazed pasture area	1000ha
QLNpas	Non-pasture vegetation area	1000ha
QLCro	Cropland area	1000ha
QLFrs	Forest area	1000ha
QLPrimf	Primary forest area	1000ha
QLSecdf	Secondary forest area	1000ha
QLNat	Other natural land area	1000ha
QLAfr	Afforested land area	1000ha
QLCroCul	Harvested cropland area	1000ha
QLCroFal	Fallow cropland area	1000ha
QL	Land allocated to a production activity	1000ha
PLPas	Pasture land price (area-weighted average)	$/ha
PLPasGrz	Grazed pasture land price	$/ha
PLPasUng	Ungrazed pasture land price	$/ha
PLNpas	Non-pasture land price	$/ha
PLCro	Cropland price	$/ha
PLFrs	Forest price	$/ha
PLAfr	Afforested land price	$/ha
PLNat	Other natural land price	$/ha
PLPrimf	Primary forest land price	$/ha
PLSecdf	Secondary forest land price	$/ha
PLCroCul	Harvested cropland price	$/ha

Continued on next page

Table 3: List of endogenous variables (continued)

Variable	Description	Unit
PL	Land price	\$/ha
PLEne	Energy crop land price	\$/ha
GHG	GHG emissions from agriculture sector	$ktCO_2$-eq/yr
GHGLU	GHG emissions from land-use sector	$ktCO_2$-eq/yr
RRed	Emission reduction rate	-
CAbt	Abatement cost	\$/t
CEmis	Unit residual-emission charge	\$/t
MuBasMin	Complementary variable for basin production	-
MuImpMin	Complementary variable for imports	-
MuExpMax	Complementary variable for the export limit	-
MuNtrdMin	Complementary variable for minimum non-traded production	-
MuPrimfCap	Complementary variable for primary forest expansion	-
MuSecdfCap	Complementary variable for secondary forest expansion	-
MuAfrCap	Complementary variable for the annual afforestation limit	-
MuPrimfProt	Complementary variable for protected primary forest	-
MuSecdfProt	Complementary variable for protected secondary forest	-
MuNatProt	Complementary variable for protected other natural land	-
MuPasProt	Complementary variable for protected pasture	-
MuCroCap	Complementary variable for the cropland constraint	-

3.3 Exogenous Variables

Table 4 lists the exogenous variables used in the model. For data sources, see Section 5.

Table 4: List of exogenous variables

Variable	Description	Data Source
qdFood0	Food consumption demand in the base year	FAOSTAT
qdBio0	Biofuel demand in the base year	FAOSTAT
qdSeed0	Seed demand in the base year	FAOSTAT
qdTour0	Tourist consumption demand in the base year	FAOSTAT
qdLoss0	Food loss in the base year	FAOSTAT
qdOth0	Other demand in the base year	FAOSTAT

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Table 4: List of exogenous variables (continued)

Variable	Description	Data Source
qs0	Domestic production in the base year	FAOSTAT
qsBas0	Basin production in the base year	FAOSTAT, GAEZ v4
ql0, pl0	Activity land area and rent in the base year	Model calibration
qdEne	Demand for dedicated energy crops	AIM-Hub
qStock	Stock change plus the base-year statistical calibration adjustment	FAOSTAT FBS
qRes	Derived residual of the global international trade balance	FAOSTAT FBS
rLoss	Food loss rate	FAOSTAT
rWaste	Food waste rate	FAO (2011)
gdpPc	GDP per capita	SSP 3.0 [6]
gdpPc0	GDP per capita in the base year	World Bank
pop	Population	SSP 3.0 [7]
pop0	Population in the base year	World Bank
mmj	Market margin (zero in the current calibration)	Model assumption
rPpSub	Production subsidy rate	GTAP 10
rPpTax	Factor employment tax rate	GTAP 10
rPcSub	Net subsidy rate on household purchases	GTAP 10
rPcSub0	Household purchase subsidy rate in the base year	GTAP 10
tm	Import tariff rate	GTAP 10
mmm	Import margin rate	GTAP 10
te	Export tariff rate	GTAP 10
mme	Export margin rate	GTAP 10
frtM, frtE	Import and export freight cost	Base-year margins and world price

Continued on next page

Table 4: List of exogenous variables (continued)

Variable	Description	Data Source
yld0	Crop yield in the base year	FAOSTAT, GAEZ v4; H08 for energy crops
yld	Crop yield before the land-rent response	Model calculation
yldTech	Yield change factor due to technological progress	IMPACT
yldChg	Yield change factor due to climate change impacts	GAEZ v4
yldPlum	Optional basin-level yield adjustment from spatial crop allocation	AIM-PLUM
croInt	Cropping intensity	MAPSPAM
yGrass	Grass yield	GAEZ v4
rAnimalProd	Slaughtering / milking rate	FAOSTAT
feedReq	Feed requirement per unit of livestock output	Herrero et al. (2018)
shrFeedConc	Share of concentrate feed	Herrero et al. (2018)
shrFeedGrain	Composition of concentrate feed	FAOSTAT
shrSys	Annual livestock system share	Model calculation
feedCoef	Annual feed input coefficient	Model calculation
animalWt	Livestock body weight	FAOSTAT
nAnimalRes	Number of non-productive livestock	FAOSTAT
coBas0	Unit cost of other production inputs by basin in the base year	FAOSTAT, GTAP 10
coCty0	Unit cost of other production inputs by country in the base year	FAOSTAT, GTAP 10
qlTot	Total land area	LUH3
qlOth	Ice/water area	LUH3
qlUrb	Urban area	LUH3; optional SEDAC extension

Continued on next page

Table 4: List of exogenous variables (continued)

Variable	Description	Data Source
`qlProt`	Protected land area by land-use class	LUH3, WDPA
`plFrs0`	Forest land price in the base year	LUH3, GTAP 10
`plPrimf0`	Primary forest land price in the base year	LUH3, GTAP 10
`plSecdf0`	Secondary forest land price in the base year	LUH3, GTAP 10
`plNatNet`	Price of other natural land before the protection constraint	LUH3, GTAP 10
`plPrimfCar`, `plSecdfCar`	Annualized carbon credit in forest land rent	AIM-PLUM, VISIT
`rPrimfExp`, `rSecdfExp`	Forest expansion limit rates	Model calibration
`afrPot`	Afforestation potential admitted by the model	AIM-PLUM
`plAfrRaw`	Afforested land price before the annual planting constraint	AIM-PLUM, VISIT
`plCroFal0`	Fallow land price in the base year	LUH3, GTAP 10
`carbonStock`	Forest carbon stock by production unit	AIM-PLUM, VISIT
`carbonStockNat`	Carbon stock of other natural land by production unit	AIM-PLUM, VISIT
`carbonSeq`	Carbon sequestration by production unit and forest age	AIM-PLUM, VISIT
`ef`	GHG emission factor in agriculture sector	FAOSTAT
`efLUCro`	Land-use GHG emission factor for cropland	FAOSTAT
`efLUFrs`	Land-use GHG emission factor for forest area	FAOSTAT
`efLUPas`	Land-use GHG emission factor for pasture area	FAOSTAT

Continued on next page

Table 4: List of exogenous variables (continued)

Variable	Description	Data Source
`efLUNat`	Land-use GHG emission factor for other natural land	FAOSTAT
`shrCEmis`	Share of the residual-emission charge applied	Model calculation
`pCar`	Carbon price	AIM-Hub
`mrp`	Maximum reduction potential of abatement technologies	Harmsen et al. (2019)

3.4 Parameters

Table 5 lists the parameters used in the model.

Table 5: List of parameters

Parameter	Description	Notes
`elaInc`	Income elasticity of consumption in the current model year	Derived from the Food Demand Model estimates. See Section .
`elaP`	Price elasticity of consumption	Estimated by regression based on Muhammad et al. (2011)
`intCoef`	Intermediate input coefficient	
`elaCO`	Elasticity of other production factors	
`elaYldPL`	Elasticity of crop yield to land rent	
δ	Coefficient parameter of the logit function	
ρ	Exponent parameter of the logit function	Land-use exponents (`ela*`) follow the nested-logit structure of GCAM [8]; production-system exponents are estimated from Herrero et al. (2018)
`resElaP`	Price-elasticity regression residual	
`shrWat`	Share of primary-crop production technologies	
`shrCz`	Climate zone share	
`rGrassUse`	Grass (pasture) utilization rate (base-year pasture split)	
`pInt0`	Calibrated base-year processing input price	USD/t

Continued on next page

Table 5: List of parameters (continued)

Parameter	Description	Notes
pFeed0	Base-year feed price	USD/t
feedCostRes	Additive feed-cost calibration residual	USD/t of livestock output
pbRes	Residual between the domestic-good market price and border-equivalent price	
pmRes	Residual between import price and world price	
peRes	Residual between export price and world price	
exr	Exchange rate	
dcffrs	Discount factor for forest carbon value	Introduces the discounted carbon value of forests, reflecting a 60-year forest lifetime and a 5% interest rate.
rFrsCarbonMax	Maximum forest carbon credit relative to base-year rent	
afrPlantMax	Annual planting limit as a share of base-year non-pasture area	
afrCost	Afforestation cost	Cost of establishing new forest land, assumed at 250 (100 US$/acre; 1 acre = 0.4 ha) based on Gorte R.W. (2009).
qlAfrAge	Afforested area by age class	Records annual afforestation by stand age.
σ	Elasticity of adoption of abatement technologies	Estimated by regression based on Harmsen et al. (2019)

4 System of Equations

This section presents the mathematical formulation of the model and its explanation. In the following, **endogenous variables** begin with uppercase letters, while **parameters** begin with lowercase letters or use Greek notation. Lowercase parameters include calibrated coefficients and exogenous or predetermined inputs.

4.1 Demand Block

This block describes the equations related to demand determination. In the demand block, food demand is determined by factors such as prices, population, and GDP.

Total Demand

$$QD_{c,cty} = QDFood_{c,cty} + QDFed_{c,cty} + QDInt_{c,cty} + QDBio_{c,cty} + QDSeed_{c,cty} + QDTour_{c,cty} + QDLoss_{c,cty} + QDWaste_{c,cty} + QDOth_{c,cty} \tag{QDEq}$$

Consumer Price Index

$$PCIdx_{c,cty} = \frac{(1 - rPcSub_{c,cty})\, PC_{c,cty}}{(1 - rPcSub0_{c,cty})\, pc0_{c,cty}} \tag{PCIdxEq}$$

Household Consumption Demand

$$QDFood_{c,cty} = qdFood0_{c,cty} \cdot \left(\frac{gdpPc_{cty}}{gdpPc0_{cty}}\right)^{elaInc_{c,cty}} \cdot \prod_{c2} PCIdx_{c2,cty}^{elaP_{c,c2,cty}} \cdot \frac{pop_{cty}}{pop0_{cty}} \tag{QDFoodEq}$$

Feed Demand

$$QDFed_{c_fed,cty} = \sum_{bas} \sum_{a_liv} QFeed_{a_liv,c_fed,bas,cty} \tag{QDFedEq}$$

Intermediate Input Demand for Processing

$$QDInt_{c_int,cty} = \sum_{a_pro} QInt_{a_pro,c_int,cty} \tag{QDIntEq}$$

Biofuel Demand

$$QDBio_{c,cty} = qdBio0_{c,cty} \cdot \prod_{c2} PCIdx_{c2,cty}^{elaP_{c,c2,cty}} \tag{QDBioEq}$$

Seed Demand

$$QDSeed_{c,cty} = qdSeed0_{c,cty} \cdot \left(\frac{gdpPc_{cty}}{gdpPc0_{cty}}\right)^{elaInc_{c,cty}} \cdot \frac{pop_{cty}}{pop0_{cty}} \cdot \prod_{c2} PCIdx_{c2,cty}^{elaP_{c,c2,cty}} \tag{QDSeedEq}$$

Tourist Consumption Demand

$$QDTour_{c,cty} = qdTour0_{c,cty} \cdot \left(\frac{gdpPc_{cty}}{gdpPc0_{cty}}\right)^{elaInc_{c,cty}} \cdot \frac{pop_{cty}}{pop0_{cty}} \cdot \prod_{c2} PCIdx_{c2,cty}^{elaP_{c,c2,cty}} \tag{QDTourEq}$$

Loss and Waste

$$QDLoss_{c,cty} = QD_{c,cty} \cdot rLoss_{c,cty} \tag{QDLossEq}$$

$$QDWaste_{c,cty} = QDFood_{c,cty} \cdot rWaste_{c,cty} \tag{QDWasteEq}$$

Other Demand

$$QDOth_{c,cty} = qdOth0_{c,cty} \cdot \left(\frac{gdpPc_{cty}}{gdpPc0_{cty}}\right)^{elaInc_{c,cty}} \cdot \frac{pop_{cty}}{pop0_{cty}} \cdot \prod_{c2} PCIdx_{c2,cty}^{elaP_{c,c2,cty}} \tag{QDOthEq}$$

4.2 Production Block

The production block describes the equations determining the production quantities of each food commodity, the input of intermediate goods and production factors, and the producer prices. The production functions are represented by Leontief-type functions, where each good is produced through the combination of specific input factors.

4.2.1 Annual Driver Calculations

Crop yields, livestock system shares, and feed input coefficients are calculated before each annual or sub-step equilibrium solve. They are therefore time-varying parameters rather than MCP equilibrium variables:

$$
\begin{aligned}
yld_{a_cro,bas,cty,wat} &= yld0_{a_cro,bas,cty,wat} \cdot yldTech_{a_cro,bas,cty,wat} \\
&\quad \cdot yldChg_{a_cro,bas,cty,wat} \cdot yldPlum_{a_cro,bas,cty}, \\
shrSys_{a_liv_rum,bas,cty,cz,\mathrm{LG}} &= \frac{1}{1+\delta^{sys}_{a_liv_rum,bas,cty,cz,\mathrm{LG}} \cdot gdpPc_{cty}^{-\rho^{sys}_{a_liv_rum,\mathrm{LG}}}}, \\
shrSys_{a_liv_rum,bas,cty,cz,\mathrm{MX}} &= 1 - shrSys_{a_liv_rum,bas,cty,cz,\mathrm{LG}}, \\
feedCoef_{a_liv,c_fed,bas,cty,cz,sys} &= feedReq_{a_liv,bas,cty,cz,sys} \cdot shrFeedConc_{a_liv,bas,cty,cz,sys} \\
&\quad \cdot shrFeedGrain_{a_liv,c_fed,bas,cty,cz,sys}.
\end{aligned}
$$

The optional $yldPlum$ term is one when AIM-PLUM feedback is not used. The LG/MX split applies to ruminants; non-ruminants use the aggregate system.

Domestic Production

$$
QS_{a_bas,cty} = \sum_{bas} QSBas_{a_bas,bas,cty} \tag{QSAggEq}
$$

4.2.2 Crop Production

Primary crops are differentiated by production technologies (irrigated and rainfed). Yield improvements due to technological progress and climate impacts are considered for crops. Crop yields also respond to land rents.

Crop Production by Technology

$$
QSWat_{a_cro,bas,cty,wat} = QSBas_{a_cro,bas,cty} \cdot shrWat_{a_cro,bas,cty,wat} \tag{QSWatEq}
$$

Producer Price of Crops

$$
\begin{aligned}
&PP_{a_cro,cty}\,(1 + rPpSub_{a_cro,cty} - rPpTax_{a_cro,cty}) \\
&= \left(PL_{a_cro,bas,cty} \cdot \left(\frac{PL_{a_cro,bas,cty}}{pl0_{a_cro,bas,cty}}\right)^{-elaYldPL} \cdot \sum_{wat} \frac{shrWat_{a_cro,bas,cty,wat}}{yld_{a_cro,bas,cty,wat} \cdot croInt_{bas,cty,a_cro,wat}} \right. \\
&\qquad \left. + \; COBas_{a_cro,bas,cty}\right) / \,(1 + MuBasMin_{a_cro,bas,cty})
\end{aligned} \tag{CroZeroProfitEq}
$$

Land Input for Crop Production

$$QL_{a_cro,bas,cty} = QSBas_{a_cro,bas,cty} \cdot \left(\frac{PL_{a_cro,bas,cty}}{pl0_{a_cro,bas,cty}}\right)^{-elaYldPL} \cdot \sum_{wat} \frac{shrWat_{a_cro,bas,cty,wat}}{yld_{a_cro,bas,cty,wat} \cdot croInt_{bas,cty,a_cro,wat}} \quad \text{(CroLandEq)}$$

Other Production Factor Price (Crops)

$$COBas_{a_cro,bas,cty} = coBas0_{a_cro,bas,cty} \cdot \left(\frac{QL_{a_cro,bas,cty}}{ql0_{a_cro,bas,cty}}\right)^{elaCO_{a_cro,cty}} \quad \text{(COCroEq)}$$

4.2.3 Livestock Production

The livestock production block describes equations for determining livestock output, production systems, feed requirements, and associated prices. Livestock products are distinguished by climate zones and production systems (pasture-based or mixed). Feed use and production efficiency depend on feed composition, feed requirements, and animal productivity.

Livestock Production by System

$$QSSys_{a_liv,bas,cty,cz,sys} = QSBas_{a_liv,bas,cty} \cdot shrCz_{a_liv,bas,cty,cz} \cdot shrSys_{a_liv,bas,cty,cz,sys} \quad \text{(QSSysEq)}$$

Producer Price of Livestock

$$\begin{aligned} PP_{a_liv,cty}\,(1 + rPpSub_{a_liv,cty} - rPpTax_{a_liv,cty}) &= \left(\frac{PL_{a_liv,bas,cty}}{YLiv_{a_liv,bas,cty}} + COBas_{a_liv,bas,cty} + feedCostRes_{a_liv,bas,cty} \right. \\ &\quad \left. + \sum_{c_fed} pFeed0_{a_liv,c_fed,cty} \frac{PC_{c_fed,cty}}{pc0_{c_fed,cty}} \right) \\ &\quad \cdot \sum_{cz,sys} \left(shrCz_{a_liv,bas,cty,cz}\, shrSys_{a_liv,bas,cty,cz,sys}\, feedCoef_{a_liv,c_fed,bas,cty,cz,sys}\right)\Bigg) \\ &\quad /(1 + MuBasMin_{a_liv,bas,cty}) \quad \text{(LivZeroProfitEq)} \end{aligned}$$

Grass Input Calculation

$$\begin{aligned} QGrass_{a_liv,bas,cty} = QSBas_{a_liv,bas,cty} &\cdot \sum_{cz,sys} \Big(shrCz_{a_liv,bas,cty,cz} \cdot shrSys_{a_liv,bas,cty,cz,sys} \\ &\cdot feedReq_{a_liv,bas,cty,cz,sys} \cdot (1 - shrFeedConc_{a_liv,bas,cty,cz,sys})\Big) \end{aligned} \quad \text{(QGrassEq)}$$

Available Grassland Area

$$QGrass_{a_liv,bas,cty} = QL_{a_liv,bas,cty} \cdot yGrass_{bas,cty} \tag{GrassBalEq}$$

Livestock Yield

$$YLiv_{a_liv,bas,cty} = \frac{QSBas_{a_liv,bas,cty}}{QL_{a_liv,bas,cty}} \tag{YLivEq}$$

Feed Input Calculation

$$\begin{aligned} QFeed_{a_liv,c_fed,bas,cty} = QSBas_{a_liv,bas,cty} \\ \cdot \sum_{cz,sys} \left(shrCz_{a_liv,bas,cty,cz} \cdot shrSys_{a_liv,bas,cty,cz,sys} \right. \\ \left. \cdot feedCoef_{a_liv,c_fed,bas,cty,cz,sys} \right) \end{aligned} \tag{QFeedEq}$$

Other Production Factor Price (Livestock)

$$COBas_{a_liv,bas,cty} = coBas0_{a_liv,bas,cty} \left(\frac{QL_{a_liv,bas,cty}}{ql0_{a_liv,bas,cty}} \right)^{elaCO_{a_liv,cty}} \tag{COLivEq}$$

The land-cost term applies only to grazing livestock. For livestock without land input, the cost ratio in Eq. (COLivEq) is $QSBas/qsBas0$.

Livestock Production Quantity

$$\begin{aligned} QSBas_{a_liv,bas,cty} = animalWt_{a_liv,bas,cty} \cdot (NAnimal_{a_liv,bas,cty} - nAnimalRes_{a_liv,bas,cty}) \\ \cdot rAnimalProd_{a_liv,cty} \end{aligned} \tag{NAnimalEq}$$

4.2.4 Processed Goods Production

This section describes the production equations for processed agricultural products. Processed products are produced using intermediate inputs and other production factors.

Producer Price of Processed Products

$$\begin{aligned} PP_{a_pro,cty} \left(1 + rPpSub_{a_pro,cty} - rPpTax_{a_pro,cty}\right) \\ = COCty_{a_pro,cty} \\ + \sum_{c_int} pInt0_{a_pro,c_int,cty} \frac{PC_{c_int,cty}}{pc0_{c_int,cty}} intCoef_{a_pro,c_int,cty} \end{aligned} \tag{ProZeroProfitEq}$$

Intermediate Input Demand for Processed Products

$$QInt_{a_pro,c_int,cty} = QS_{a_pro,cty} \cdot intCoef_{a_pro,c_int,cty} \tag{QIntEq}$$

Other Production Factor Price (Processed Products)

$$COCty_{a_pro,cty} = coCty0_{a_pro,cty} \cdot \left(\frac{QS_{a_pro,cty}}{qs0_{a_pro,cty}} \right)^{elaCO_{a_pro,cty}} \tag{COProEq}$$

4.2.5 Energy Crop Production

Energy crop production is determined from regional totals provided exogenously by the AIM-Hub model [3].

Regional Energy Crop Supply Balance

$$qdEne_{reg} = \sum_{bas,cty \in reg} QSBas_{\text{ecr},bas,cty} \tag{EneBalEq}$$

Land Input for Energy Crops

$$QL_{\text{ecr},bas,cty} = \frac{QSBas_{\text{ecr},bas,cty}}{yld_{\text{ecr},bas,cty,\text{Rainfed}}} \tag{EneLandEq}$$

Land Price for Energy Crops

$$PL_{\text{ecr},bas,cty} = PLEne_{reg} \tag{PLEneEq}$$

4.3 Price Block

This subsection describes the equations in the price block. The price block represents the relationships among various prices in the model. Prices are linked through fixed coefficients so that the price used in the demand and supply functions is consistently determined.

Consumer and Producer Prices For non-traded goods,

$$PC_{c,cty} = \frac{(1 + mmj_{c,cty})\, PP_{c,cty} + CAbt_{c,cty} + CEmis_{c,cty}}{1 + MuNtrdMin_{c,cty}} \tag{PCNtrdEq}$$

For traded goods, the domestic-good price satisfies

$$\begin{aligned} PB_{c,cty}\,(1 + pbRes_{c,cty}) = (1 + mmj_{c,cty})\, PP_{c,cty} \\ + CAbt_{c,cty} + CEmis_{c,cty} \end{aligned} \tag{PBEq}$$

Consumer and Border Prices Under the price-parity trade formulation, the consumer price is weighted by base-year domestic supply and imports:

$$PC_{c,cty} = PB_{c,cty}\,(1 + pbRes_{c,cty}) + \left[PM_{c,cty} - PB_{c,cty}\,(1 + pbRes_{c,cty})\right] \frac{qm0_{c,cty}}{qs0_{c,cty} - qe0_{c,cty} + qm0_{c,cty}} \quad \text{(PCTrdEq)}$$

The import term is zero where base-year imports are zero.

Import Price

$$PM_{c,cty} = (PW_c + frtM_{c,cty})\,exr_{cty} \cdot (1 + tm_{c,cty} + MuImpMin_{c,cty})(1 + pmRes_{c,cty}) \quad \text{(PMEq)}$$

Export Price For the price-parity trade formulation,

$$PE_{c,cty} = \frac{(PW_c - frtE_{c,cty})\,(1 + peRes_{c,cty})\,exr_{cty}}{1 + te_{c,cty} + MuExpMax_{c,cty}} \quad \text{(PEEq)}$$

4.4 Land Use Block

The land-use block allocates available land among different uses through a Logit function. Each nest uses a common exponent. Non-pasture vegetation is allocated among cropland, forest, other natural land, and afforested land. Protected areas are imposed as lower bounds for the corresponding land classes, and the agro-ecological suitability constraint applies to cropland. Urban land is assigned exogenously. In the default configuration, the LUH3-based base-year urban area is held constant over future years; as an optional extension, SSP-specific urban projections from the SEDAC urban dataset can be used to update urban land over time.

Vegetation Area

$$QLVeg_{bas,cty} = qlTot_{bas,cty} - qlOth_{bas,cty} - qlUrb_{bas,cty} \quad \text{(QLVegBalEq)}$$

Pasture Area Allocation

$$QLPas_{bas,cty} = QLVeg_{bas,cty} \cdot \frac{\delta^{\text{pas}}_{bas,cty} \cdot PLPas^{\rho^{\text{pas}}_{bas,cty}}_{bas,cty}}{\delta^{\text{pas}}_{bas,cty} \cdot PLPas^{\rho^{\text{pas}}_{bas,cty}}_{bas,cty} + \delta^{\text{npas}}_{bas,cty} \cdot PLNpas^{\rho^{\text{npas}}_{bas,cty}}_{bas,cty}} \quad \text{(PasLogitEq)}$$

Grazed and Ungrazed Pasture Allocation

$$QLPasGrz_{bas,cty} = QLPas_{bas,cty} \cdot \frac{\delta^{\text{pasgrz}}_{bas,cty} \cdot PLPasGrz^{\rho^{\text{pasgrz}}_{bas,cty}}_{bas,cty}}{\delta^{\text{pasgrz}}_{bas,cty} \cdot PLPasGrz^{\rho^{\text{pasgrz}}_{bas,cty}}_{bas,cty} + \delta^{\text{pasung}}_{bas,cty} \cdot PLPasUng^{\rho^{\text{pasung}}_{bas,cty}}_{bas,cty}} \quad \text{(PasGrzLogitEq)}$$

$$QLPasUng_{bas,cty} = QLPas_{bas,cty} \cdot \frac{\delta^{\text{pasung}}_{bas,cty} \cdot PLPasUng^{\rho^{\text{pasung}}_{bas,cty}}_{bas,cty}}{\delta^{\text{pasgrz}}_{bas,cty} \cdot PLPasGrz^{\rho^{\text{pasgrz}}_{bas,cty}}_{bas,cty} + \delta^{\text{pasung}}_{bas,cty} \cdot PLPasUng^{\rho^{\text{pasung}}_{bas,cty}}_{bas,cty}} \quad \text{(PasUngLogitEq)}$$

Non-pasture Vegetation Area

$$QLNpas_{bas,cty} = QLVeg_{bas,cty} \cdot \frac{\delta^{\text{npas}}_{bas,cty} PLNpas^{\rho^{\text{npas}}_{bas,cty}}_{bas,cty}}{\delta^{\text{pas}}_{bas,cty} PLPas^{\rho^{\text{pas}}_{bas,cty}}_{bas,cty} + \delta^{\text{npas}}_{bas,cty} PLNpas^{\rho^{\text{npas}}_{bas,cty}}_{bas,cty}} \quad \text{(NpasLogitEq)}$$

Cropland Area

$$QLCro_{bas,cty} = QLNpas_{bas,cty} \cdot \frac{\delta^{\text{cro}}_{bas,cty} PLCro^{\rho^{\text{cro}}_{bas,cty}}_{bas,cty}}{\delta^{\text{cro}}_{bas,cty} PLCro^{\rho^{\text{cro}}_{bas,cty}}_{bas,cty} + \delta^{\text{frs}}_{bas,cty} PLFrs^{\rho^{\text{frs}}_{bas,cty}}_{bas,cty} + \delta^{\text{nat}}_{bas,cty} PLNat^{\rho^{\text{nat}}_{bas,cty}}_{bas,cty} + \delta^{\text{afr}}_{bas,cty} PLAfr^{\rho^{\text{afr}}_{bas,cty}}_{bas,cty}} \quad \text{(CroLogitEq)}$$

Forest Area

$$QLFrs_{bas,cty} = QLNpas_{bas,cty} \cdot \frac{\delta^{\text{frs}}_{bas,cty} PLFrs^{\rho^{\text{frs}}_{bas,cty}}_{bas,cty}}{\delta^{\text{cro}}_{bas,cty} PLCro^{\rho^{\text{cro}}_{bas,cty}}_{bas,cty} + \delta^{\text{frs}}_{bas,cty} PLFrs^{\rho^{\text{frs}}_{bas,cty}}_{bas,cty} + \delta^{\text{nat}}_{bas,cty} PLNat^{\rho^{\text{nat}}_{bas,cty}}_{bas,cty} + \delta^{\text{afr}}_{bas,cty} PLAfr^{\rho^{\text{afr}}_{bas,cty}}_{bas,cty}} \quad \text{(FrsLogitEq)}$$

Primary and Secondary Forest Allocation

$$QLPrimf_{bas,cty} = QLFrs_{bas,cty} \cdot \frac{\delta^{\text{primf}}_{bas,cty} PLPrimf^{\rho^{\text{primf}}_{bas,cty}}_{bas,cty}}{\delta^{\text{primf}}_{bas,cty} PLPrimf^{\rho^{\text{primf}}_{bas,cty}}_{bas,cty} + \delta^{\text{secdf}}_{bas,cty} PLSecdf^{\rho^{\text{secdf}}_{bas,cty}}_{bas,cty}} \quad \text{(PrimfLogitEq)}$$

$$QLSecdf_{bas,cty} = QLFrs_{bas,cty} \cdot \frac{\delta^{\text{secdf}}_{bas,cty} PLSecdf^{\rho^{\text{secdf}}_{bas,cty}}_{bas,cty}}{\delta^{\text{primf}}_{bas,cty} PLPrimf^{\rho^{\text{primf}}_{bas,cty}}_{bas,cty} + \delta^{\text{secdf}}_{bas,cty} PLSecdf^{\rho^{\text{secdf}}_{bas,cty}}_{bas,cty}} \quad \text{(SecdfLogitEq)}$$

Other Natural Land

$$QLNat_{bas,cty} = QLNpas_{bas,cty} \cdot \frac{\delta^{\text{nat}}_{bas,cty} PLNat^{\rho^{\text{nat}}_{bas,cty}}_{bas,cty}}{\delta^{\text{cro}}_{bas,cty} PLCro^{\rho^{\text{cro}}_{bas,cty}}_{bas,cty} + \delta^{\text{frs}}_{bas,cty} PLFrs^{\rho^{\text{frs}}_{bas,cty}}_{bas,cty} + \delta^{\text{nat}}_{bas,cty} PLNat^{\rho^{\text{nat}}_{bas,cty}}_{bas,cty} + \delta^{\text{afr}}_{bas,cty} PLAfr^{\rho^{\text{afr}}_{bas,cty}}_{bas,cty}} \quad \text{(NatLogitEq)}$$

Afforested Land

$$QLAfr_{bas,cty} = QLNpas_{bas,cty} \cdot \frac{\delta^{\text{afr}}_{bas,cty} PLAfr^{\rho^{\text{afr}}_{bas,cty}}_{bas,cty}}{\delta^{\text{cro}}_{bas,cty} PLCro^{\rho^{\text{cro}}_{bas,cty}}_{bas,cty} + \delta^{\text{frs}}_{bas,cty} PLFrs^{\rho^{\text{frs}}_{bas,cty}}_{bas,cty} + \delta^{\text{nat}}_{bas,cty} PLNat^{\rho^{\text{nat}}_{bas,cty}}_{bas,cty} + \delta^{\text{afr}}_{bas,cty} PLAfr^{\rho^{\text{afr}}_{bas,cty}}_{bas,cty}} \quad \text{(AfrLogitEq)}$$

Harvested Cropland Area

$$QLCroCul_{bas,cty} = QLCro_{bas,cty} \cdot \frac{\delta^{\text{crocul}}_{bas,cty} PLCroCul^{\rho^{\text{crocul}}_{bas,cty}}_{bas,cty}}{\delta^{\text{crocul}}_{bas,cty} PLCroCul^{\rho^{\text{crocul}}_{bas,cty}}_{bas,cty} + \delta^{\text{crofal}}_{bas,cty} PLCroFal^{\rho^{\text{crofal}}_{bas,cty}}_{bas,cty}} \quad \text{(CroCulLogitEq)}$$

Fallow Cropland Area

$$QLCroFal_{bas,cty} = QLCro_{bas,cty} \cdot \frac{\delta^{\text{crofal}}_{bas,cty} PLCroFal^{\rho^{\text{crofal}}_{bas,cty}}_{bas,cty}}{\delta^{\text{crocul}}_{bas,cty} PLCroCul^{\rho^{\text{crocul}}_{bas,cty}}_{bas,cty} + \delta^{\text{crofal}}_{bas,cty} PLCroFal^{\rho^{\text{crofal}}_{bas,cty}}_{bas,cty}} \quad \text{(CroFalLogitEq)}$$

Cropland Allocation

$$QL_{a_cro,bas,cty} = QLCroCul_{bas,cty} \cdot \frac{\delta_{a_cro,bas,cty} \cdot PL_{a_cro,bas,cty}^{\rho_{a_cro,bas,cty}}}{\sum_{a_cro2} \delta_{a_cro2,bas,cty} \cdot PL_{a_cro2,bas,cty}^{\rho_{a_cro2,bas,cty}}} \quad \text{(CroDistEq)}$$

Pasture Allocation

$$QL_{a_liv_rum,bas,cty} = QLPasGrz_{bas,cty} \cdot \frac{\delta_{a_liv_rum,bas,cty} \cdot PL_{a_liv_rum,bas,cty}^{\rho_{a_liv_rum,bas,cty}}}{\sum_{a_liv_rum2} \delta_{a_liv_rum2,bas,cty} \cdot PL_{a_liv_rum2,bas,cty}^{\rho_{a_liv_rum2,bas,cty}}} \quad \text{(LivDistEq)}$$

Land Prices The price drivers for forest, other natural land, and afforestation include their land-carbon value before the associated complementarity wedges are applied.

$$PLCroCul_{bas,cty} \cdot QLCroCul_{bas,cty} = \sum_{a_cro} PL_{a_cro,bas,cty} \cdot QL_{a_cro,bas,cty} \quad \text{(PLCroCulEq)}$$

$$PLCro_{bas,cty} \cdot QLCro_{bas,cty} = \left(PLCroCul_{bas,cty} \cdot QLCroCul_{bas,cty} + PLCroFal_{bas,cty} \cdot QLCroFal_{bas,cty}\right) \quad /(1 + MuCroCap_{bas,cty}) \quad \text{(PLCroEq)}$$

$$PLNpas_{bas,cty}^{\rho_{bas,cty}^{\text{npas}}} \cdot qlNpas0_{bas,cty} = plNpas0_{bas,cty}^{\rho_{bas,cty}^{\text{npas}}} \cdot \left(\delta^{\text{cro}} PLCro^{\rho^{\text{cro}}} + \delta^{\text{frs}} PLFrs^{\rho^{\text{frs}}} + \delta^{\text{nat}} PLNat^{\rho^{\text{nat}}} + \delta^{\text{afr}} PLAfr^{\rho^{\text{afr}}}\right)_{bas,cty} \quad \text{(PLNpasEq)}$$

$$PLPasGrz_{bas,cty} \cdot QLPasGrz_{bas,cty} = \sum_{a_liv_rum} PL_{a_liv_rum,bas,cty} \cdot QL_{a_liv_rum,bas,cty} \quad \text{(PLPasGrzEq)}$$

$$PLPas_{bas,cty} \cdot QLPas_{bas,cty} = \left(PLPasGrz_{bas,cty} \cdot QLPasGrz_{bas,cty} + PLPasUng_{bas,cty} \cdot QLPasUng_{bas,cty}\right) \cdot (1 + MuPasProt_{bas,cty}) \quad \text{(PLPasEq)}$$

$$PLFrs_{bas,cty}^{\rho_{bas,cty}^{\text{frs}}} \cdot qlFrs0_{bas,cty} = plFrs0_{bas,cty}^{\rho_{bas,cty}^{\text{frs}}} \cdot \left(\delta^{\text{primf}} PLPrimf^{\rho^{\text{primf}}} + \delta^{\text{secdf}} PLSecdf^{\rho^{\text{secdf}}}\right)_{bas,cty} \quad \text{(PLFrsEq)}$$

$$PLPrimf_{bas,cty} = (plPrimf0_{bas,cty} + plPrimfCar_{bas,cty}) \cdot \frac{1 + MuPrimfProt_{bas,cty}}{1 + MuPrimfCap_{bas,cty}} \quad \text{(PLPrimfEq)}$$

$$PLSecdf_{bas,cty} = (plSecdf0_{bas,cty} + plSecdfCar_{bas,cty}) \cdot \frac{1 + MuSecdfProt_{bas,cty}}{1 + MuSecdfCap_{bas,cty}} \quad \text{(PLSecdfEq)}$$

The annualized carbon credit for primary forest is

$$plPrimfCar_{bas,cty} = \min\Big\{rFrsCarbonMax\, plPrimf0_{bas,cty}, \max\Big[0, \left(carbonStock_{bas,cty} \tfrac{44}{12} pCarLand_{bas,cty} - afrCost_{cty}\right) dcffrs_{cty}\Big]\Big\}.$$

Secondary forest uses the same formula with $plSecdf0$. A credit is zero after the protected-area floor of that class has bound.

$$PLNat_{bas,cty} = plNatNet_{bas,cty} \cdot (1 + MuNatProt_{bas,cty}) \quad \text{(PLNatEq)}$$

Afforestation Price and Annual Limit The rent driver $plAfrRaw$ is annualized carbon revenue over the forest rotation less annualized establishment cost.

$$PLAfr_{bas,cty} = \frac{plAfrRaw_{bas,cty}}{1 + MuAfrCap_{bas,cty}} \quad \text{(PLAfrEq)}$$

$$QLAfr_{bas,cty} \le \min\Big\{qlAfrPrev_{bas,cty} + afrPlantMax \cdot qlNpas0_{bas,cty},\ afrPot_{bas,cty}\Big\} \quad \perp \quad MuAfrCap_{bas,cty} \ge 0 \quad \text{(AfrCapEq)}$$

Forest Expansion Constraints

$$QLPrimf_{bas,cty} \le qlPrimfPrev_{bas,cty} \cdot (1 + rPrimfExp_{bas,cty}) \quad \perp \quad MuPrimfCap_{bas,cty} \ge 0 \quad \text{(PrimfCapEq)}$$

$$QLSecdf_{bas,cty} \leq qlSecdfPrev_{bas,cty} \cdot (1 + rSecdfExp_{bas,cty}) \perp MuSecdfCap_{bas,cty} \geq 0 \quad \text{(SecdfCapEq)}$$

Protected-Area Constraints Protected areas are floors within the corresponding gross land areas; pasture and rangeland share one pasture floor.

$$QLPrimf_{bas,cty} \geq qlProt_{bas,cty,\text{primf}} \perp MuPrimfProt_{bas,cty} \geq 0, \quad \text{(PrimfProtEq)}$$

$$QLSecdf_{bas,cty} \geq qlProt_{bas,cty,\text{secdf}} \perp MuSecdfProt_{bas,cty} \geq 0, \quad \text{(SecdfProtEq)}$$

$$QLNat_{bas,cty} \geq qlProt_{bas,cty,\text{gra}} \perp MuNatProt_{bas,cty} \geq 0, \quad \text{(NatProtEq)}$$

$$QLPas_{bas,cty} \geq \sum_{lu_pas} qlProt_{bas,cty,lu_pas} \perp MuPasProt_{bas,cty} \geq 0. \quad \text{(PasProtEq)}$$

Cropland Constraint

$$qlCroCap_{bas,cty} \geq QLCro_{bas,cty} \perp MuCroCap_{bas,cty} \geq 0 \quad \text{(CroCapEq)}$$

4.5 Market Block

This subsection describes the equations in the market block, where consistency between food demand and supply is ensured. Trade of food commodities is represented as a complementarity problem. When the import price is higher than the border price, the import quantity becomes zero. Once the import price decreases to equal the border price, imports begin. Similarly, when the border price is higher than the export price, exports become zero, and once the export price rises to match the border price, exports take place.

Domestic Trade Balance

$$NT_{c,cty} = QS_{c,cty} - QD_{c,cty} - qStock_{c,cty} \quad \text{(ComBalEq)}$$

$$NT_{c,cty} = QE_{c,cty} - QM_{c,cty} \quad \text{(NTEq)}$$

International Trade Balance

$$\sum_{cty} NT_{c,cty} - qRes_c = 0 \quad \text{(WldBalEq)}$$

Import and Export Conditions

$$PM_{c,cty} \geq PB_{c,cty} \quad \perp \quad QM_{c,cty} \geq 0 \tag{ImpParityEq}$$

$$PB_{c,cty} \geq PE_{c,cty} \quad \perp \quad QE_{c,cty} \geq 0 \tag{ExpParityEq}$$

4.6 GHG Emission Block

This subsection describes the equations in the GHG emission block. GHG emissions are calculated based on the production quantities of each food commodity. When carbon prices are considered, the amount of mitigation technology adoption and its cost are computed from the MAC (Marginal Abatement Cost) curve. Emissions from the land-use sector are calculated according to each land area.

GHG Emissions

$$GHG_{a,src,gas,cty} = QS_{a,cty} \cdot ef_{a,src,gas,cty} \cdot (1 - RRed_{a,src,gas,cty}) \tag{GHGAgEq}$$

Land-Use Emissions

$$\begin{aligned} GHGLU_{src,gas,cty} = \sum_{bas} \Big(& efLUCro_{src,gas,cty} \cdot QLCro_{bas,cty} \\ & + efLUFrs_{src,gas,cty} \cdot QLFrs_{bas,cty} \\ & - carbonStock_{bas,cty} \cdot \frac{44}{12} \cdot \min\left\{0,\ QLFrs_{bas,cty} - qlFrsPrev_{bas,cty}\right\} \\ & + efLUPas_{src,gas,cty} \cdot QLPas_{bas,cty} + efLUNat_{src,gas,cty} \cdot QLNat_{bas,cty} \Big) \end{aligned} \tag{GHGLUEq}$$

$$GHGLU_{"afr","CO_2",cty} = -\sum_{bas} \sum_{age} qlAfrAge_{bas,cty,age} \cdot carbonSeq_{bas,cty,age} \tag{AfrGHGEq}$$

Emission Reduction Rate

$$RRed_{a,src,gas,cty} = mrp_{src,gas,cty} \cdot \left(1 - \left[1/(1 + pCar_{cty})\right]^{\sigma_{src,gas,cty}}\right) \tag{RRedEq}$$

Abatement Cost

$$\begin{aligned} CAbt_{a,cty} = \sum_{src,gas} ef_{a,src,gas,cty} \cdot \Bigg[& \frac{\sigma_{src,gas,cty}}{(\sigma_{src,gas,cty} - 1)} \cdot mrp_{src,gas,cty} \\ & \cdot \left(1 - \left(1 - RRed_{a,src,gas,cty}/mrp_{src,gas,cty}\right)^{(1-1/\sigma_{src,gas,cty})}\right) - RRed_{a,src,gas,cty} \Bigg] \end{aligned} \tag{CAbtEq}$$

Carbon Cost on Remaining Emissions The factor $shrCEmis$ limits the charge relative to the previous consumer price.

$$CEmis_{a,cty} = shrCEmis_{a,cty} \sum_{src,gas} pCar_{cty} \cdot ef_{a,src,gas,cty} \cdot (1 - RRed_{a,src,gas,cty}) \quad \text{(CEmisEq)}$$

4.7 Complementarity Conditions

This subsection describes the complementarity conditions introduced to avoid numerical errors in the Logit function when production quantities in specific production units become zero. If the production of a unit becomes zero, its land-use area would also become zero, leading to instability. Therefore, a minimum positive value is imposed to ensure that production quantities do not fall to zero.

Minimum Production Share for Trade

$$\frac{QS_{c,cty}}{QM_{c,cty}} \geq 0.01 \quad \perp \quad MuImpMin_{c,cty} \geq 0 \quad \text{(ImpMinEq)}$$

Minimum Production Share for Supply

$$\frac{QSBas_{a,bas,cty}}{QS_{a,cty}} \geq 0.01 \quad \perp \quad MuBasMin_{a,bas,cty} \geq 0 \quad \text{(QSBasMinEq)}$$

5 Data and Preprocessing

This section explains the calibration and processing of data used in the model. Exogenous variables and parameters were estimated using multiple datasets, adjusted to obtain equilibrium in the base year, and parameters were estimated by regression analysis.

5.1 Data Sources

Table 6 lists the datasets used in the model.

Table 6: List of data sources

Data Source	Dataset Name	URL
FAOSTAT [9]	Food Balances	https://www.fao.org/faostat/en/#data/FBS
FAOSTAT [9]	Production	https://www.fao.org/faostat/en/#data/QCL
FAOSTAT [9]	Trade	https://www.fao.org/faostat/en/#data/TCL
FAOSTAT [9]	Producer Prices	https://www.fao.org/faostat/en/#data/PP
FAOSTAT [9]	Land Use	https://www.fao.org/faostat/en/#data/RL
FAOSTAT [9]	Emissions	https://www.fao.org/faostat/en/#data/GCE
FAO Land and Water Division [2]	Major Hydrological Basins of the World	https://data.apps.fao.org/catalog//iso/77070 86d-af3c-41cc-8aa5-323 d8609b2d1
FAO and IIASA [10]	GAEZ v4	https://gaez.fao.org/
IIASA	SSP 3.0	https://data.ece.iiasa.ac.at/ssp/#/about
GTAP [11]	GTAP 10	https://jgea.org/ojs/index.php/jgea/article/view/77
IFPRI [12]	MAPSPAM	https://doi.org/10.7910/DVN/SWPENT
IFPRI [13]	IMPACT	https://github.com/IFPRI/IMPACT
World Bank	World Bank Open Data	https://data.worldbank.org/
EDGAR [14]	EDGAR v8.0	https://edgar.jrc.ec.europa.eu/dataset_ghg80
Herrero et al. (2018) [15]	Livestock production systems	https://doi.org/10.4225/08/5aa068b33fe06
Muhammad et al. (2011) [16]	International Evidence on Food Consumption Patterns	http://199.135.94.241/publications/pub-details/?pubid=47581
T. P. Robinson et al. (2018) [17]	Global distribution of ruminant livestock production systems V5	https://doi.org/10.7910/DVN/WPDSZE

Continued on next page

Table 6: List of data sources (continued)

Data Source	Dataset Name	URL
Harmsen et al. (2019) [18]	CH_4 and N_2O MAC curves	https://doi.org/10.1016/j.envsci.2019.05.013
Hurtt et al. (2019) [19]	Land Use Harmonization 2	https://luh.umd.edu/index.shtml
LUH	Land Use Harmonization 3	https://luh.umd.edu/
Hanasaki et al. (2008) [20, 21]	H08 rainfed crop yield	https://doi.org/10.5194/hess-12-1007-2008
Hasegawa et al. (2017) [5]	AIM-PLUM spatial land-use outputs	https://doi.org/10.1016/j.scitotenv.2016.12.025
Ito and Inatomi (2012) [22]	VISIT carbon-cycle outputs	https://doi.org/10.1175/JHM-D-10-05034.1
Gao and Pesaresi (2021) [23]	Global 1-km Downscaled Urban Land Extent Projection and Base Year Grids by SSP Scenarios, 2000-2100	https://doi.org/10.7927/1z4r-ez63
Gao and Pesaresi (2021) [24]	Downscaling SSP-consistent Global Spatial Urban Land Projections from 1/8-degree to 1-km Resolution, 2000-2100	https://doi.org/10.1038/s41597-021-01052-0
IUCN and UNEP-WCMC [25]	World Database on Protected Areas (WDPA), Dec 2025 release	https://www.protectedplanet.net

5.2 Calibration and processing of data

5.2.1 Adjustment of demand-side data

First, the adjustment of demand-side data was conducted. In the model, if a country–commodity pair has missing data for the base year 2015, it is assumed that consumption does not occur thereafter. Therefore, for country–commodity pairs without 2015 data in the Food Balance Sheets (FBS), the average value of subsequent years was used as the 2015 consumption level.

$$qdFood_{(c,cty,2015)} = \frac{\sum_{yr} qdFood^{FAO}_{(c,cty,yr)}}{N_{yr}} \quad \text{for } qdFood_{(c,cty,2015)} = \text{NA} \tag{5.1}$$

where $qdFood^{FAO}$ is household consumption data from FAO, and N_{yr} is the number of years for which data exist.

In the FBS data, household consumption includes the amount of food wasted at the household level. To distinguish this, waste rates by seven commodities and seven regions were obtained from FAO (2011) [26]. Using these coefficients, household consumption and food waste were estimated as follows:

$$qdWaste_{(c,cty)} = qdFood^{FAO}_{(c,cty)} \cdot rWaste_{(c,cty)} \tag{5.2}$$

$$qdFood_{(c,cty)} = qdFood^{FAO}_{(c,cty)} \cdot (1 - rWaste_{(c,cty)}) \tag{5.3}$$

For model simplification, the commodities used as livestock feed, processing materials, and biofuel inputs were limited. Wheat, rice, maize, other cereals, roots and tubers, sugar crops, pulses, oil crops, vegetables, and oilseed cakes were used as livestock feed. Wheat, rice, maize, other cereals, and fruits were used as raw materials for alcoholic beverages. Sugar crops were used as inputs for sugars, oil crops for vegetable oils and oilseed cakes, and maize, sugar crops, and vegetable oils for biofuels. Feed demand, intermediate input demand, and biofuel demand for other commodities were set to zero.

$$qdFed_{(c \notin c_{fed},cty)} = 0 \tag{5.4}$$

$$qdBio_{(c \notin c_{bio},cty)} = 0 \tag{5.5}$$

$$qdInt_{(c \notin c_{int},cty)} = 0 \tag{5.6}$$

Dairy quantities are expressed in raw-milk equivalent, and dairy unit values use the same quantity basis.

The FBS do not cover oilseed cakes, so their supply and utilization were taken from the FAOSTAT Supply Utilization Accounts. Crushing yields vegetable oil and oilseed cake jointly. The crushed oil crops were split between the two products in proportion to their output values at world export unit values:

$$intCoef_{(a,ocr,cty)} = \frac{qdInt_{(ocr,cty)} \cdot peWld_{(a)}}{\sum_{a' \in a_{crush}} \left(peWld_{(a')} \cdot qs_{(a',cty)}\right)} \quad a \in a_{crush} = \{vol, cak\} \tag{5.7}$$

where $peWld$ is the export-weighted world unit value of each product.

As a result of these adjustments, the total domestic demand changed. If the sum of all demand components was less than the total demand from the data, the residual was added as “other demand”:

$$\begin{aligned} qdOth_{(c,cty)} = qd_{(c,cty)} - \\ \big(&qdFood_{(c,cty)} + qdFed_{(c,cty)} + qdInt_{(c,cty)} + qdSeed_{(c,cty)} + \\ &qdTour_{(c,cty)} + qdLoss_{(c,cty)} + qdWaste_{(c,cty)}\big) \end{aligned} \tag{5.8}$$

5.2.2 Adjustment of crop production data

Next, the adjustment of crop production data was conducted. From the FBS data, the production quantities of primary crops for each country were obtained. These were disaggregated into production units and production technologies using the GAEZ data as follows:

$$qsWat_{(a_cro,bas,cty,wat)} = qs^{FAO}_{(a_cro,cty)} \cdot \frac{qsGaez^{GAEZ}_{(a_cro,bas,cty,wat)}}{\sum_{(bas,wat)} qsGaez^{GAEZ}_{(a_cro,bas,cty,wat)}} \tag{5.9}$$

where qs^{FAO} denotes production quantity data from FAO, and $qsGaez^{GAEZ}$ represents production data from GAEZ.

National harvested and fallow cropland totals were derived from LUH3 cropland and FAOSTAT land-use shares. Harvested land was allocated among crops using national FAOSTAT harvested areas, and then among basins and technologies using crop-specific GAEZ shares:

$$qlCroCul_{cty} = \sum_{bas} qlCro^{LUH3}_{(bas,cty)} \frac{qlCroCul^{FAO}_{cty}}{qlCroCul^{FAO}_{cty} + qlCroFal^{FAO}_{cty}}, \tag{5.10}$$

$$qlWat_{(a_cro,bas,cty,wat)} = qlCroCul_{cty} \frac{qlHarvFao_{(a_cro,cty)}}{\sum_{a_cro2} qlHarvFao_{(a_cro2,cty)}} \cdot \frac{qlGaez^{GAEZ}_{(a_cro,bas,cty,wat)}}{\sum_{bas,wat} qlGaez^{GAEZ}_{(a_cro,bas,cty,wat)}}. \tag{5.11}$$

Here, $qlHarvFao$ is FAOSTAT harvested area and $qlGaez$ is GAEZ physical crop area. Basin harvested cropland is the sum of $qlWat$ over crops and technologies; fallow cropland follows the national FAOSTAT fallow share.

From the MAPSPAM dataset, harvested and physical area data were obtained for each crop, production unit, and production technology. Using these data, the cropping intensity was estimated as follows:

$$croInt_{(bas,cty,a_cro,wat)} = \frac{mapspam^{Harvested}_{(a_cro,bas,cty,wat)}}{mapspam^{Physical}_{(a_cro,bas,cty,wat)}} \tag{5.12}$$

where $mapspam^{Harvested}$ is the harvested area data from MAPSPAM, and $mapspam^{Physical}$ is the physical area data from MAPSPAM.

Crop yields were calculated using the obtained production quantities, harvested areas, and cropping intensities as follows:

$$yld_{(a_cro,bas,cty,wat)} = \frac{qsWat_{(a_cro,bas,cty,wat)}}{qlWat_{(a_cro,bas,cty,wat)} \cdot croInt_{(bas,cty,a_cro,wat)}} \tag{5.13}$$

Food-crop yields are based on FAOSTAT production and GAEZ spatial allocation. For dedicated energy crops, AIM-ALPHA instead uses the rainfed yield from H08, following the bioenergy-potential assessment of Wu et al. [20, 21, 27].

When feedback from AIM-PLUM is enabled, the crop-yield driver is adjusted by the change in basin-average yield associated with changes in the grid-level spatial allocation of crop production [5]. The adjustment is one when this feedback is not used.

5.2.3 Adjustment of livestock production data

Next, the adjustment of livestock production data was conducted. From FAO data, the numbers of livestock animals and slaughtered (or milked) animals were obtained. Using these, the production rate was calculated. A minimum threshold of 0.3 was set for the production rate; if it fell below this value, it was assumed that a portion of livestock was not involved in production activities.

$$rAnimalProd_{(a_liv,cty)} = \frac{nAnimal^{FAO}_{(a_liv,cty)}}{nAnimalStock^{FAO}_{(a_liv,cty)}} \tag{5.14}$$

$$nAnimalRes_{(a_liv,cty)} = nAnimalStock^{FAO}_{(a_liv,cty)} - \frac{nAnimal^{FAO}_{(a_liv,cty)}}{0.3} \quad (\text{if } rAnimalProd_{(a_liv,cty)} < 0.3) \tag{5.15}$$

where $nAnimal^{FAO}$ is the number of slaughtered (or milked) animals from FAO, and $nAnimalStock^{FAO}$ is the total livestock population.

When non-productive livestock exist, the corresponding unused pasture area was estimated as:

$$qlPasRes_{(bas,cty)} = qlPas^{LUH3}_{(bas,cty)} \cdot \frac{\sum_{a_liv_rum} nAnimalRes_{(a_liv_rum,cty)}}{\sum_{a_liv_rum} nAnimalStock^{FAO}_{(a_liv_rum,cty)}} \tag{5.16}$$

where $qlPasRes$ denotes the unused pasture area, and $qlPas^{LUH3}$ is the pasture area from LUH3.

Country-level livestock production data from the FBS were disaggregated into production units, climate zones, and production systems using the dataset from Herrero et al. (2018). For ruminants, the share of each production unit is its share of the national pasture area. Climate-zone shares are obtained from Herrero et al. (2018), and system shares use the GDP-based regression described in Section 6.

$$qsSys_{(a_liv,bas,cty,cz,sys)} = qs^{FAO}_{(a_liv,cty)} \cdot shrBas_{(a_liv,bas,cty)} \cdot shrCz_{(a_liv,bas,cty,cz)} \cdot shrSys_{(a_liv,bas,cty,cz,sys)}. \tag{5.17}$$

$$qlSys_{(a_liv,bas,cty,cz,sys)} = \left(qlPas^{LUH3}_{(bas,cty)} - qlPasRes_{(bas,cty)}\right) \cdot rGrassUse_{(bas,cty)} \cdot \frac{qGrassSys_{(a_liv,bas,cty,cz,sys)}}{\sum_{a_liv2,cz2,sys2} qGrassSys_{(a_liv2,bas,cty,cz2,sys2)}}. \tag{5.18}$$

Pasture is allocated in proportion to calibrated grass demand. The grass utilization rate $rGrassUse$ is calculated from estimated grass demand and available supply.

The grazed and ungrazed pasture areas are then computed as:

$$qlPasGrz_{(bas,cty)} = \sum_{(a_liv,cz,sys)} qlSys_{(a_liv,bas,cty,cz,sys)} \tag{5.19}$$

$$qlPasUng_{(bas,cty)} = qlPas^{LUH3}_{(bas,cty)} - qlPasGrz_{(bas,cty)} \tag{5.20}$$

Next, the feed requirements and concentrate feed share were estimated as follows:

$$feedReq_{(a_liv,bas,cty,cz,sys)} = \frac{feedQty^{HERRERO}_{(a_liv,bas,cty,cz,sys)} + pasArea^{HERRERO}_{(a_liv,bas,cty,cz,sys)}}{livWt^{HERRERO}_{(a_liv,bas,cty,cz,sys)}} \tag{5.21}$$

$$shrFeedConc_{(a_liv,bas,cty,cz,sys)} = \frac{feedQty^{HERRERO}_{(a_liv,bas,cty,cz,sys)}}{feedQty^{HERRERO}_{(a_liv,bas,cty,cz,sys)} + pasArea^{HERRERO}_{(a_liv,bas,cty,cz,sys)}} \tag{5.22}$$

where $feedQty^{HERRERO}$ is the amount of concentrate feed from Herrero et al. (2018).

Using these values and the FBS feed demand data, concentrate feed use was adjusted with a country-level correction coefficient $feedScale$, based on the ratio of national feed demand to estimated concentrate feed requirements. Feed commodities followed their shares in national feed use:

$$\begin{aligned} qFeedSys_{(a_liv,c_fed,bas,cty,cz,sys)} = {} & qsSys_{(a_liv,bas,cty,cz,sys)} \\ & \cdot feedReq_{(a_liv,bas,cty,cz,sys)} \cdot shrFeedConc_{(a_liv,bas,cty,cz,sys)} \\ & \cdot feedScale_{cty} \cdot shrFeedGrain_{(c_fed,cty)}. \end{aligned} \tag{5.23}$$

Grass demand is

$$\begin{aligned} qGrassSys_{(a_liv,bas,cty,cz,sys)} = {} & qsSys_{(a_liv,bas,cty,cz,sys)} \\ & \cdot feedReq_{(a_liv,bas,cty,cz,sys)} \cdot \left(1 - shrFeedConc_{(a_liv,bas,cty,cz,sys)}\right). \end{aligned} \tag{5.24}$$

If estimated grass demand exceeded available supply, it was reduced proportionally. Feed requirements, concentrate feed shares, and grass utilization were recalculated from the adjusted quantities.

5.2.4 Adjustment of food trade

Next, the adjustment of food trade data was conducted. In order to express trade through complementary conditions, the net import and net export quantities were derived as follows:

$$qm_{(c,cty)} = qm^{FAO}_{(c,cty)} - qe^{FAO}_{(c,cty)} \quad (\text{if } qm^{FAO}_{(c,cty)} > qe^{FAO}_{(c,cty)}) \tag{5.25}$$

$$qe_{(c,cty)} = qe^{FAO}_{(c,cty)} - qm^{FAO}_{(c,cty)} \quad (\text{if } qe^{FAO}_{(c,cty)} > qm^{FAO}_{(c,cty)}) \tag{5.26}$$

where qm^{FAO} denotes the import quantity data from FAO, and qe^{FAO} denotes the export quantity data from FAO.

In the FBS dataset, global trade is not balanced; that is, total world exports and imports do not match. The discrepancy is retained as the signed accounting residual $qRes$, rather than interpreted as a physical transport loss.

$$qRes_c = \sum_{cty} qe_{(c,cty)} - \sum_{cty} qm_{(c,cty)} \tag{5.27}$$

5.2.5 Estimation of land constraints

Next, the estimation method for land areas unsuitable for agricultural use due to soil and topographical constraints is described. From the GAEZ dataset, area data for 57 agro-ecological zone (AEZ) classes, classified according to climate, soil, and topographical conditions, were obtained. Among these, land classes with strong constraints were selected and considered unsuitable for agricultural use. The cropland upper bound was obtained from the remaining suitable share:

$$qlCroCap_{(bas,cty)} = qlTot^{LUH3}_{(bas,cty)} \cdot \left(1 - \frac{\sum_{aezlim} qlGaez_{(aezlim,bas,cty)}}{\sum_{aez} qlGaez_{(aez,bas,cty)}}\right) \tag{5.28}$$

where:

- $qlTot^{LUH3}$: total land area from LUH3,
- $qlGaez$: area data by AEZ class from GAEZ,
- aez: the set of 57 AEZ classes,
- $aezlim \in aez$: AEZ classes with severe land-use constraints.

In addition, protected-area information was incorporated using the World Database on Protected Areas (WDPA) [25]. The WDPA polygons were overlaid with basin-country units and classified into primary forest, secondary forest, other natural land, pasture, and rangeland. These areas form $qlProt_{(bas,cty,lu_prot)}$, which is imposed as a lower bound on the corresponding land class; pasture and rangeland share the pasture constraint.

Urban land is based on the LUH3 base-year area in the default model configuration. The base-year value is held constant in future years unless the optional SEDAC urban-projection extension is enabled. For this extension, the SEDAC 1-km urban fraction data [23, 24] were aggregated to basin-country units and converted to annual SSP-specific series. When the extension is switched on, these SEDAC series are used

to update urban land in scenario runs while preserving the LUH3-based base-year level. To maintain physical consistency of the land balance, urban area was bounded by zero and by available land after subtracting water/ice.

5.2.6 Estimation of forest carbon data

Forest and other natural-land carbon stocks are derived from spatial outputs of AIM-PLUM and VISIT and aggregated to basin-country units [5, 22]. Forest growth by stand age is used to calculate carbon sequestration from annual afforestation [28].

5.2.7 Adjustment of price data

From the FAO dataset, producer prices, import prices, and export prices for each country and commodity were obtained. Since the price data contained many outliers and missing values, appropriate preprocessing was conducted.

First, producer, import, and export prices outside 0.1–10 times the cross-country median were removed. Outlying years within each country price series were also adjusted. Processed-good producer prices were obtained from import and export unit values, weighted by traded quantities where available.

Missing producer prices were first filled using the country's relative price in the nearest observed years. Remaining gaps were filled from a quantity-weighted world price and the country's average relative price across commodities:

$$pAvg_{c,yr} = \frac{\sum_{cty \in O_{c,yr}} pp_{(c,cty,yr)}\, qs_{(c,cty,yr)}}{\sum_{cty \in O_{c,yr}} qs_{(c,cty,yr)}}, \tag{5.29}$$

$$pRelAvg_{cty,yr} = \frac{1}{N_{cty,yr}} \sum_{c \in O_{cty,yr}} \frac{pp_{(c,cty,yr)}}{pAvg_{c,yr}}, \tag{5.30}$$

$$pp_{(c,cty,yr)} = pAvg_{c,yr}\, pRelAvg_{cty,yr}. \tag{5.31}$$

Here, O denotes available observations and N their number. Import and export prices were filled similarly with the corresponding trade quantities as weights. World-average prices provide the final fallback; remaining producer-price gaps use the import-price level.

Next, production subsidies, factor employment taxes, and net subsidies on household purchases were calculated from GTAP 10. Production-side rates use output at market prices as their denominator:

$$rPpSub_{(i,cty)} = \frac{OSEP^{GTAP}_{(i,cty)} + \sum_{ii} ISEP^{GTAP}_{(ii,i,cty)} + \sum_{endw} FBEP^{GTAP}_{(endw,i,cty)}}{TVOM^{GTAP}_{(i,cty)}}, \tag{5.32}$$

$$rPpTax_{(i,cty)} = \frac{\sum_{endw} FTRV^{GTAP}_{(endw,i,cty)}}{TVOM^{GTAP}_{(i,cty)}}, \tag{5.33}$$

$$rPcSub_{(i,cty)} = 1 - \frac{VDPA^{GTAP}_{(i,cty)} + VIPA^{GTAP}_{(i,cty)}}{VDPM^{GTAP}_{(i,cty)} + VIPM^{GTAP}_{(i,cty)}}. \tag{5.34}$$

Here, $OSEP$, $ISEP$, and $FBEP$ are production, intermediate-input, and factor subsidies; $FTRV$ is factor employment tax; and $TVOM$ is output at market prices. $VDPA$, $VIPA$, $VDPM$, and $VIPM$ are domestic and imported household purchases at agent and market prices. Production costs therefore sum to $PP(1 + rPpSub - rPpTax)$, while household demand responds to $(1 - rPcSub)PC$.

Indices: i: production sector; ii: intermediate input sector; $endw$: primary factor of production; cty: reporting country; $ccty$: partner country;

The marketing margin mmj is zero in the current calibration. Trade margins mmm and mme are estimated from the CIF–FOB value gap:

$$mmm_{(i,ccty)} = \frac{\sum_{cty} VIWS^{GTAP}_{(i,cty,ccty)} - \sum_{cty} VXWD^{GTAP}_{(i,cty,ccty)}}{\sum_{cty} VXWD^{GTAP}_{(i,cty,ccty)}} \tag{5.35}$$

$$mme_{(i,cty)} = \frac{\sum_{ccty} VIWS^{GTAP}_{(i,cty,ccty)} - \sum_{ccty} VXWD^{GTAP}_{(i,cty,ccty)}}{\sum_{ccty} VIWS^{GTAP}_{(i,cty,ccty)}} \tag{5.36}$$

where:

- $VDFA^{GTAP}, VIFA^{GTAP}, EVFA^{GTAP}$: intermediate and factor input values at consumer prices,
- $VIWS^{GTAP}, VXWD^{GTAP}$: import (CIF) and export (FOB) values,

The import tariff (tm) and export tariff (te) were then calculated. Each tariff rate is the corresponding value ratio minus one.

$$tm_{(i,ccty)} = \frac{\sum_{cty} VIMS^{GTAP}_{(i,cty,ccty)}}{\sum_{cty} VIWS^{GTAP}_{(i,cty,ccty)}} - 1 \tag{5.37}$$

$$te_{(i,cty)} = \frac{\sum_{ccty} VXWD^{GTAP}_{(i,cty,ccty)}}{\sum_{ccty} VXMD^{GTAP}_{(i,cty,ccty)}} - 1 \tag{5.38}$$

where:

- $VIMS^{GTAP}$: import value at domestic (market) prices,
- $VXMD^{GTAP}$: export value at domestic (market) prices.

Finally, GTAP input–output data were used to separate land, feed, processing inputs, and other production costs. The cost shares use total intermediate and factor input costs as their denominator:

$$g_ipt_{(i,cty)} = \sum_{ii} \left(VDFA^{GTAP}_{(ii,i,cty)} + VIFA^{GTAP}_{(ii,i,cty)} \right) + \sum_{endw} EVFA^{GTAP}_{(endw,i,cty)}, \tag{5.39}$$

$$shrLandCost_{(i,cty)} = \frac{\sum_{ilnd} EVFA^{GTAP}_{(ilnd,i,cty)}}{g_ipt_{(i,cty)}}, \tag{5.40}$$

$$shrFeedCost_{(i,cty)} = \frac{\sum_{ifed} \left(VDFA^{GTAP}_{(ifed,i,cty)} + VIFA^{GTAP}_{(ifed,i,cty)} \right)}{g_ipt_{(i,cty)}}, \tag{5.41}$$

$$shrIntCost_{(i,cty)} = \frac{\sum_{iimd} \left(VDFA^{GTAP}_{(iimd,i,cty)} + VIFA^{GTAP}_{(iimd,i,cty)} \right)}{g_ipt_{(i,cty)}}. \tag{5.42}$$

The sets $ilnd$, $ifed$, and $iimd$ identify land/natural resources, feed, and processing inputs. Applying these shares to $pp(1 + rPpSub - rPpTax)$ gives the base-year factor costs; other production costs are the remainder.

Feed prices $pFeed0$ equal the base-year consumer prices. The additive residual $feedCostRes$ reconciles the cost of the physical feed inputs with the GTAP feed-cost share. Processing input prices $pInt0$ are scaled so that the cost of the calibrated input coefficients equals the GTAP intermediate-input cost. These input prices subsequently change in proportion to $PC/pc0$.

6 Parameter Estimation

6.1 Parameter estimation through regression analysis

This section presents the results of regression analyses used to estimate the parameters of the model. The regression equations use observed or estimated data for the dependent and independent variables.

Since price elasticities in the model were assumed to vary with per-capita GDP, the following regression equation was estimated:

$$elaP_{(c,cty)} = \alpha_c^{pe} + \beta_c^{pe} \cdot \log(gdpPc_{cty}) \tag{6.1}$$

For livestock products, it was assumed that production systems change with economic development. Thus, the parameters determining the production share were estimated using logistic regression analysis. In this estimation, dummy variables were introduced to represent climatic zones.

$$\log\left(\frac{shrSys_{(a_liv,bas,cty,cz,sys)}}{1 - shrSys_{(a_liv,bas,cty,cz,sys)}}\right) = \delta^{sys}_{(a_liv,cz,sys)} + \rho^{sys}_{(a_liv,sys)} \cdot \log(gdpPc_{cty}) \tag{6.2}$$

It was further assumed that feed requirements and the share of concentrate feed also change with the level of economic development. Regression analyses were conducted using the following equations:

$$\log\left(feedReq_{(a_liv,bas,cty,cz,sys)}\right) = \alpha^{fe}_{(a_liv,cz,sys)} + \beta^{fe}_{(a_liv,cz,sys)} \cdot \log(gdpPc_{cty}) \tag{6.3}$$

$$\log\left(\frac{shrFeedConc_{(a_liv,bas,cty,cz,sys)}}{1 - shrFeedConc_{(a_liv,bas,cty,cz,sys)}}\right) = \delta^{fs}_{(a_liv,cz,sys)} + \rho^{fs}_{(a_liv,sys)} \cdot \log(gdpPc_{cty}) \tag{6.4}$$

Following Harmsen et al. (2019), the elasticity parameter σ related to the adoption of emission-reduction technologies was estimated using the following regression equation:

$$\log\left(1 - \frac{RRed_{(src,gas,cty)}}{mrp_{(src,gas,cty)}}\right) = \sigma_{(src,gas,cty)} \cdot \log(pCar_{cty} + 1) \tag{6.5}$$

6.2 Results of parameter estimation

This section presents the results of the regression analyses conducted to estimate the model parameters.

6.2.1 Relationship between price elasticity and per-capita GDP

Figure 3 illustrates the relationship between per-capita GDP and price elasticity. Table 7 shows the estimated parameters and the coefficient of determination (R^2).

Since the model assumes that price elasticity varies with per-capita GDP, the parameters α_c^{pe} and β_c^{pe} were estimated using the regression model given in Eq. (6.1).

For all commodities, the estimated elasticities increased with higher levels of per-capita GDP. For most commodities, high coefficients of determination ($R^2 \approx 0.8$) were obtained. However, for commodities such as sugars, eggs, and spices, R^2 values were lower, around 0.5 or 0.3. The p-values for all regressions were below 0.001, indicating statistically significant relationships between the variables. Furthermore, the 95% confidence intervals for β^{pe} did not include zero, confirming the statistical significance of the estimated coefficients.

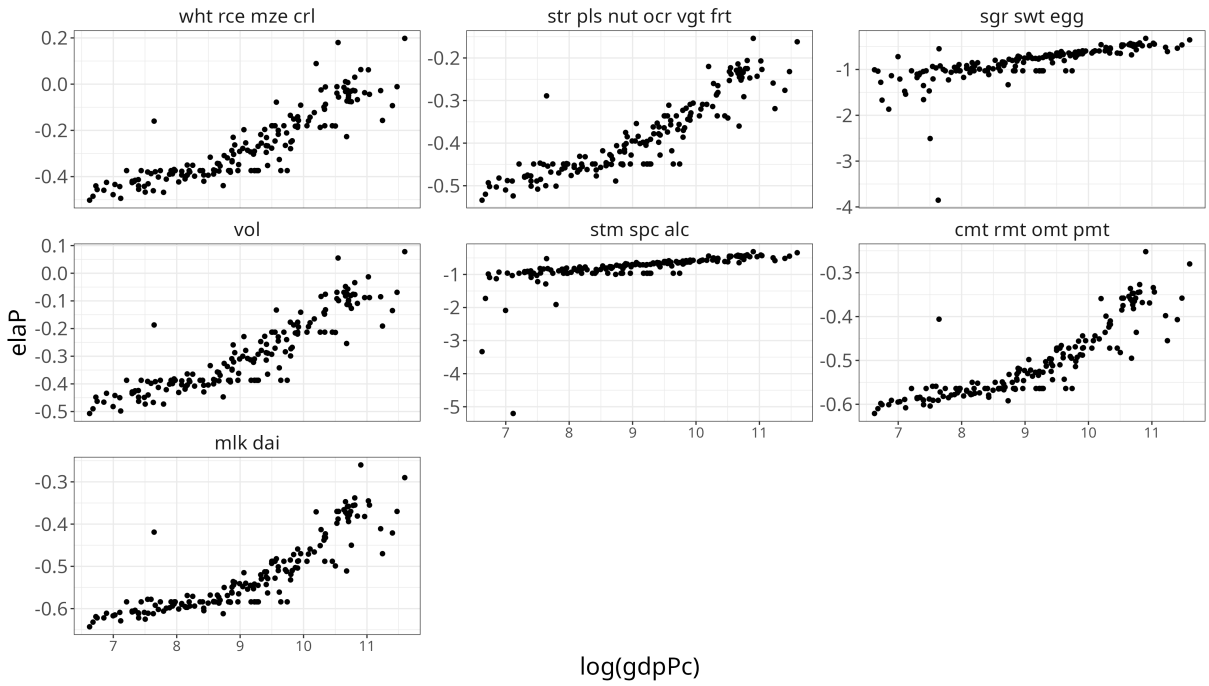


Figure 3: Relationship between per-capita GDP and price elasticity

Table 7: Estimated parameters of price elasticity from regression analysis

Commodity	α_{pe}	β_{pe}	β_{pe} (95% CI)		R^2	p-value
			lower	upper		
wht rce mze crl	−1.28	0.11	0.10	0.12	0.79	< 0.001
str pls nut ocr vgt frt	−0.99	0.07	0.06	0.07	0.81	< 0.001
sgr swt egg	−2.83	0.22	0.18	0.25	0.48	< 0.001
vol	−1.18	0.10	0.09	0.10	0.80	< 0.001
stm spc alc	−2.79	0.22	0.17	0.27	0.33	< 0.001
cmt rmt omt pmt	−1.04	0.06	0.05	0.06	0.76	< 0.001
mlk dai	−1.08	0.06	0.06	0.07	0.76	< 0.001

6.2.2 Relationship between livestock production system share and per-capita GDP

Figure 4 illustrates the relationship between per-capita GDP and the proportion of pasture-based production by livestock type and climatic zone. Table 8 presents the estimated parameters.

The parameters $\delta^{sys}_{(a_liv,cz,sys)}$ and $\rho^{sys}_{(a_liv,sys)}$ were estimated using the logistic regression model in Eq. (6.2).

As per-capita GDP increases, the share of pasture-based production decreases for all livestock products. Adjusted coefficients of determination were moderate (0.2–0.3), and all p-values were below 0.001, indicating statistically significant relationships. The 95% confidence intervals for ρ^{sys} did not include zero, confirming the significance of the coefficients.

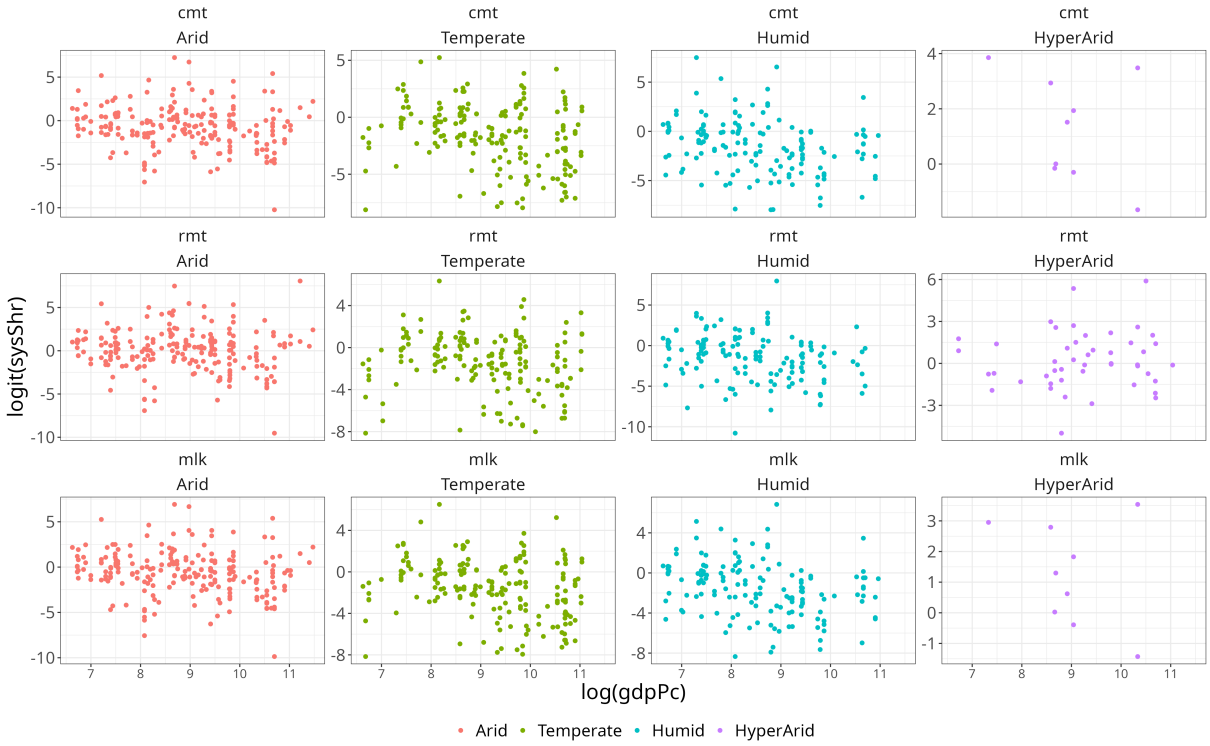


Figure 4: Relationship between per-capita GDP and share of pasture-based production

Table 8: Estimated parameters for pasture-based production share (95% confidence intervals in parentheses)

Commodity	ρ_{sys}	δ_{sys} by climate zone				R^2	p-value
		Arid	Temperate	Humid	HyperArid		
cmt	−0.51 (−0.69–−0.33)	3.87	3.07	2.86	5.88	0.24	< 0.001
rmt	−0.34 (−0.53–−0.15)	3.05	1.66	1.46	3.38	0.15	< 0.001
mlk	−0.54 (−0.71–−0.36)	4.15	3.36	3.03	6.06	0.24	< 0.001

6.2.3 Relationship between feed requirements and per-capita GDP

Figure 5 shows the relationship between per-capita GDP and feed requirements by livestock type, climate zone, and production system. Table 9 presents the estimated regression parameters, obtained using the model in Eq. (6.3).

Higher per-capita GDP is associated with lower feed requirements. For ruminants, the adjusted R^2 ranged from 0.66 to 0.97, indicating high explanatory power, while for poultry and other livestock it was relatively low (0.22 and 0.05, respectively). The coefficients β^{fe} were smaller for ruminants, suggesting stronger efficiency improvements with economic development. All p-values were below 0.001, and the 95% confidence intervals for α^{fe} did not include zero, confirming statistical significance.

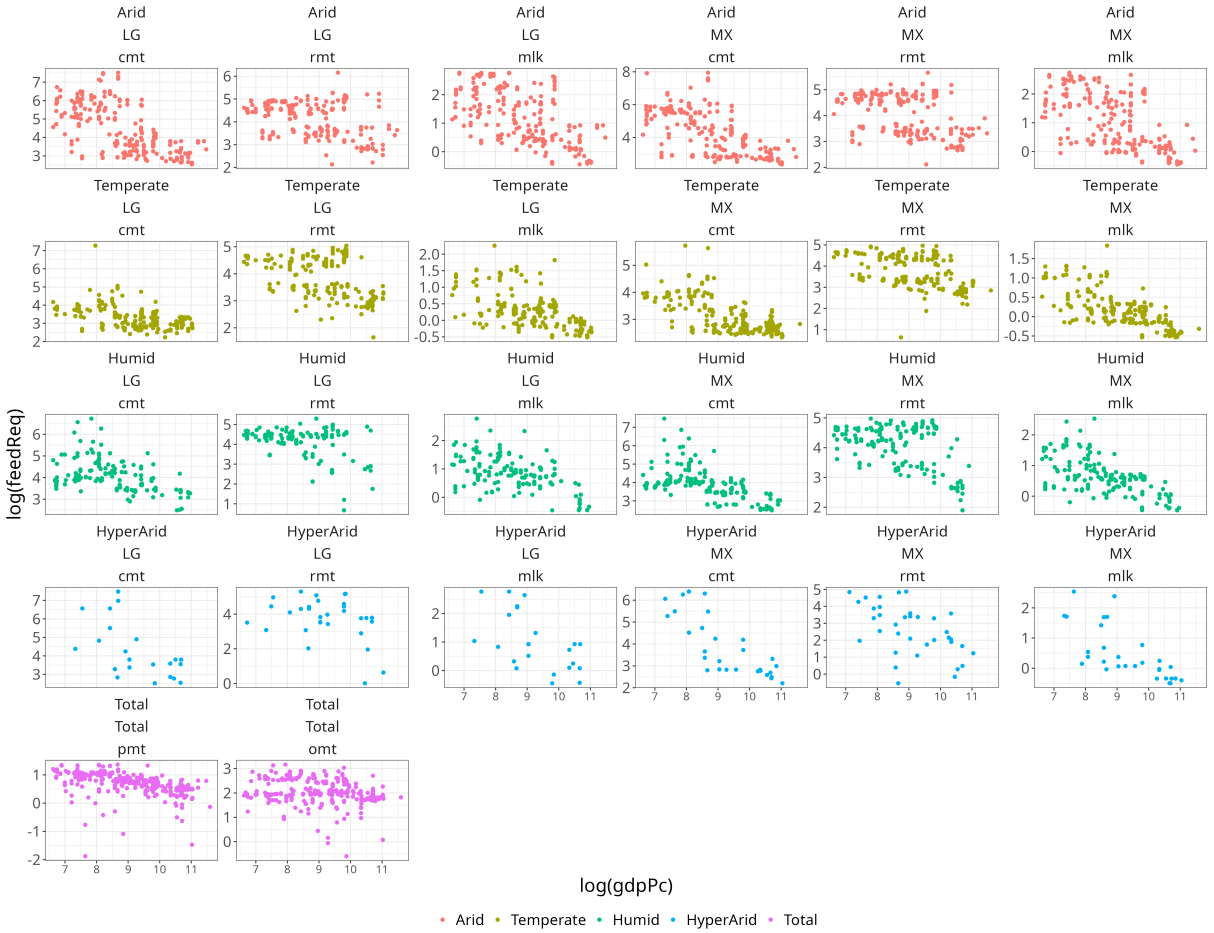


Figure 5: Relationship between per-capita GDP and feed requirements

Table 9: Estimated parameters of feed requirements (95% confidence intervals in parentheses)

Commodity	System	β_{fe}	α_{fe} by climate zone					R^2	p-value
			Arid	Temperate	Humid	HyperArid	Total		
cmt	LG	−0.47 (−0.52–−0.41)	8.58	7.65	8.07	8.80		0.96	< 0.001
rmt	LG	−0.27 (−0.32–−0.21)	6.47	6.23	6.40	6.21		0.97	< 0.001
mlk	LG	−0.37 (−0.41–−0.33)	4.45	3.69	4.01	4.49		0.71	< 0.001
cmt	MX	−0.52 (−0.57–−0.48)	8.86	7.97	8.44	8.70		0.96	< 0.001
rmt	MX	−0.33 (−0.38–−0.28)	7.01	6.76	6.96	5.61		0.97	< 0.001
mlk	MX	−0.38 (−0.41–−0.34)	4.36	3.62	3.92	3.99		0.66	< 0.001
pmt	Total	−0.14 (−0.17–−0.11)					2.02	0.21	< 0.001
omt	Total	−0.09 (−0.13–−0.04)					2.82	0.05	< 0.001

6.2.4 Relationship between concentrate feed share and per-capita GDP

Figure 6 shows the relationship between per-capita GDP and the share of concentrate feed by livestock type, climatic zone, and production system. Table 10 reports the estimated regression parameters, based on

the logistic specification in Eq. (6.4). Poultry and other livestock are excluded from the analysis because they do not consume pasture.

The results indicate that higher per-capita GDP is associated with a higher share of concentrate feed and a lower share of pasture. Adjusted coefficients of determination range from 0.74 to 0.90, indicating a relatively high explanatory power. All p-values are below 0.001, suggesting statistically significant relationships between the variables. Furthermore, the 95% confidence intervals for ρ^{fs} do not include zero, which confirms the statistical significance of the estimated coefficients.

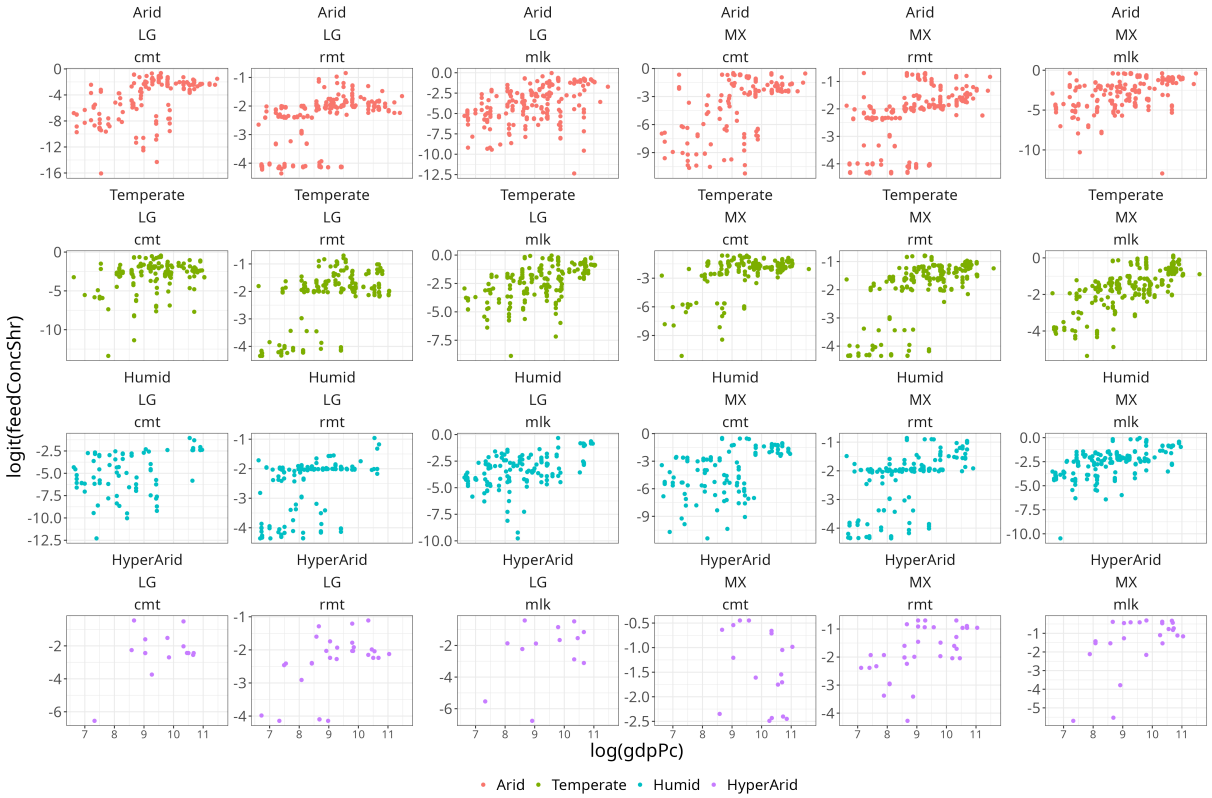


Figure 6: Relationship between per-capita GDP and concentrate feed share

Table 10: Estimated parameters of concentrate feed share (95% confidence intervals in parentheses)

Commodity	System	ρ_{fs}	δ_{fs} by climate zone				R^2	p-value
			Arid	Temperate	Humid	HyperArid		
cmt	LG	0.95 (0.74–1.16)	−13.24	−11.94	−13.09	−11.49	0.76	< 0.001
rmt	LG	0.42 (0.36–0.48)	−6.25	−6.00	−6.11	−6.36	0.90	< 0.001
mlk	LG	0.77 (0.65–0.89)	−10.68	−9.44	−9.99	−9.62	0.79	< 0.001
cmt	MX	1.04 (0.88–1.21)	−13.51	−12.21	−13.29	−11.89	0.74	< 0.001
rmt	MX	0.52 (0.46–0.57)	−6.89	−6.69	−6.81	−6.56	0.88	< 0.001
mlk	MX	0.70 (0.61–0.78)	−9.11	−8.05	−8.59	−8.06	0.78	< 0.001

7 Model Calculation Flow and Execution

7.1 Model Calculation Flow

Figure 7 shows the processing flow of the AIM-ALPHA model. The model begins with the processing of the original data sources, including extraction of necessary information, aggregation of gridded datasets, and conversion of data formats. Next, the processed data are adjusted through parameter estimation via regression analysis, data calibration to ensure equilibrium in the base year, and coefficient calibration for model consistency.

Using these prepared data, the model then performs future scenario calculations. For each year, equilibrium solutions are iteratively computed based on the base-year data, the results of the previous year, and the scenario assumptions. The aggregated annual results provide the final outputs for each scenario. Additionally, model results are converted into the submission templates of AgMIP (Agricultural Model Intercomparison and Improvement Project) and IAMC (Integrated Assessment Modeling Consortium) to facilitate data exchange and comparison. Quantities in tonnes are reported in primary equivalent: each processed good is converted into the raw materials it was made from with its intermediate input coefficient. The processing of raw materials into processed goods is therefore not reported as a use. When enabled in `model_setting.sh`, the AgMIPAssessment submodule is also executed to run vetting and variable-consistency checks, and to generate plots and summary tables.

Optional pre-run bridges convert outputs from AIM-Hub, AFOLUB, and AIM-PLUM into AIM-ALPHA inputs. The switches `use_hub_output`, `use_afolu_output`, and `use_plum_output` control these bridges. When a scenario suffix is specified by `ModelInt`, the corresponding file in `setting/modelint` defines the linked-model setting.

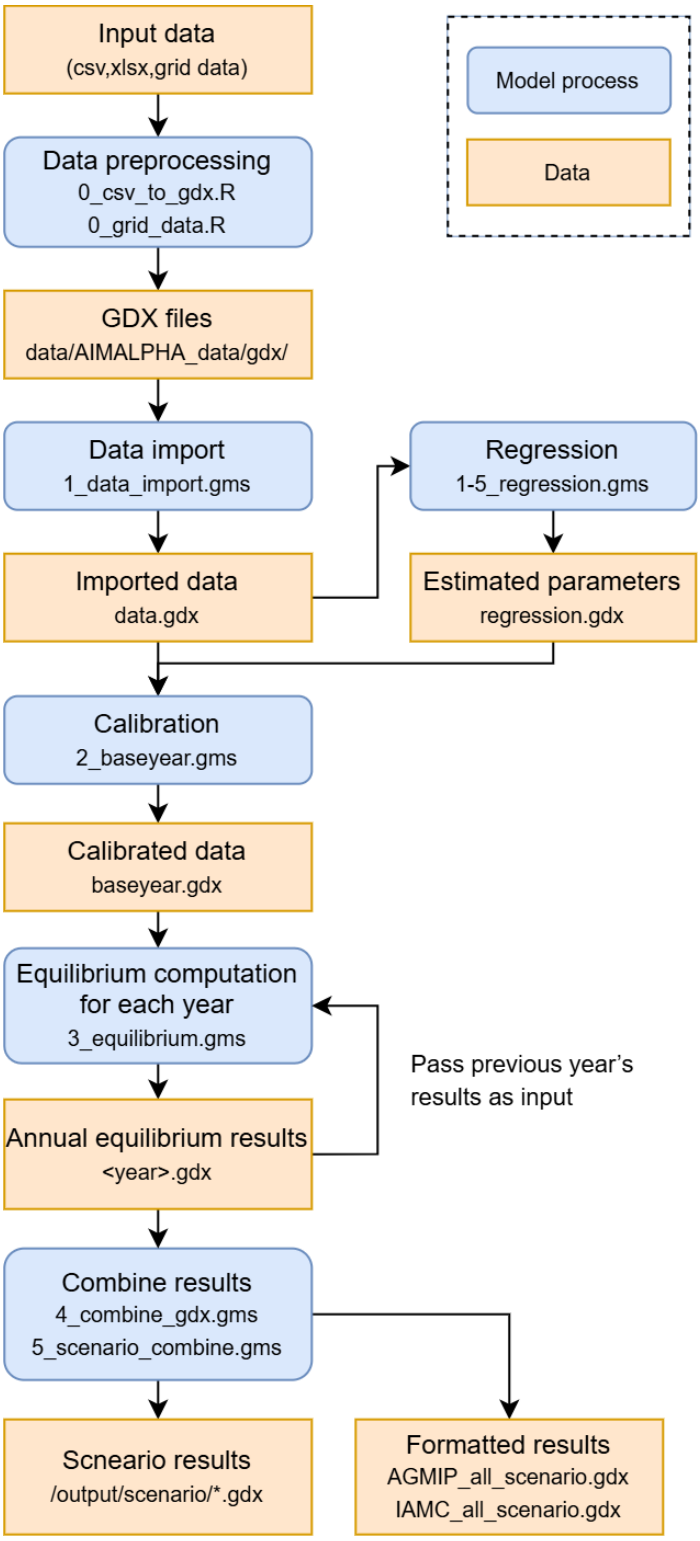


Figure 7: Processing flow of the AIM-ALPHA model

7.2 Programs within the Model

This section provides an overview of the major programs used in the AIM-ALPHA model. Each component plays a specific role in data processing, parameter calibration, and scenario simulation.

0_csv_to_gdx.R

Converts datasets in `csv` or `xlsx` format into `gdx` format. This script processes raw datasets such as FAOSTAT and MAC Curve sources.

0_grid_data.R

Aggregates gridded datasets by country and production unit, and exports the results in `gdx` format. It processes geospatial data from sources such as GAEZ, Livestock System, and LUH3.

1_data_import.gms

Integrates and reformats the `gdx`-formatted datasets into structures compatible with the model's computation framework. It performs country code conversions, consolidates data into AIM-ALPHA's food and land-use classifications, and includes submodules that output historical data in AgMIP and IAMC template formats.

1-5_regression.gms

Performs parameter regression analyses based on the original datasets to estimate elasticities and adjustment coefficients.

2_baseyear.gms

Adjusts baseline datasets to ensure equilibrium consistency in the base year.

3_equilibrium.gms

Solves for annual equilibrium states for future years, using base-year data and previous results as inputs.

4_combine_gdx.gms

Aggregates annual simulation outputs and compiles them into scenario-level result files.

5_scenario_combine.gms

Combines outputs from all scenarios to produce the comprehensive simulation dataset.

Vis_each_scenario.R

Generates graphical outputs and visualization figures from the simulation results.

tool/AgMIPAssessment/shell/execution.sh

Executes the AgMIPAssessment submodule, using `AGMIP_all_scenario.gdx` and historical templates for vetting, variable-consistency checks, and plot/table generation.

7.3 Execution Method

The AIM-ALPHA model can be executed by running the shell script `shell/ALPHA_exe.sh` on either Cygwin or Ubuntu environments.
Model configurations are defined in `shell/settings/model_setting.sh`. As summarized in Table 11, the configurable items include the selection of executable programs, scenario settings, and computational environment settings.

Basic execution

To run the model, move to the `shell` directory and execute:

```
cd shell
bash ALPHA_exe.sh
```

When `ALPHA_exe.sh` is called without any arguments, the default configuration file `model_setting.sh` is loaded automatically and the model is executed using those settings.

Using a custom setting file

It is also possible to specify an alternative setting file as an argument. In this case, the configuration defined in the specified file is used instead of `model_setting.sh`:

```
cd shell
bash ALPHA_exe.sh mysetting
```

Here, `mysetting.sh` denotes a user-defined setting file (e.g. a copy of `model_setting.sh` with modified options). Each run of `bash ALPHA_exe.sh <setting_file>` loads the corresponding setting file and executes the model according to those configurations.

Table 11: Model configuration parameters in `model_setting.sh`

Setting item	Description	Default
Model switches		
`data_download`	Executes a script to download data from submodules when the repository is first cloned. Once downloaded, this can be turned off.	on
`data_import`	Executes `1_data_import.gms` and `1-5_regression.gms`.	on
`baseyear_run`	Executes `2_baseyear.gms`.	on
`model_run`	Executes `3_equilibrium.gms`.	on
`combine_gdx`	Executes `4_combine_gdx.gms`.	on
`scenario_combine`	Executes `5_scenario_combine.gms`.	on
`Visualization`	Master switch for the plotting step; the `Vis_*` switches select what is drawn.	on
`AgMIP_assessment`	Executes AgMIPAssessment vetting and variable-consistency checks, and outputs plots and tables.	off
`csv2gdx_update`	Executes `0_csv_to_gdx.R`. If no data update is needed, this may be turned off.	off
`grid2gdx_update`	Executes `0_grid_data.R`. If no data update is needed, this may be turned off. If executed, the required gridded input files must be placed in `data`.	off
`use_hub_output`	Executes the Hub2ALPHA pre-run bridge.	off
`use_afolu_output`	Executes the AFOLU2ALPHA pre-run bridge.	off
`use_plum_output`	Executes the PLUM2ALPHA pre-run bridge.	off
`backcasting_run`	Runs the model backward from the base year to 2000.	off
`history_diagnose`	Compares the conditioned historical period with model results.	off
`surplus_run`	Calculates surplus changes relative to `surplus_reference`.	off
`report_year_interval`	Sets the reporting interval to one or five years.	5
`Vis_each_scenario`	Generates figures for individual scenarios.	on
`Vis_scenario_compare`	Enables comparison across scenarios in plots.	off
`Vis_scenario_list`	Restricts scenario-comparison plots to the listed scenarios; an empty list uses all scenarios.	empty
`Vis_model_compare`	Enables comparison across models in plots.	off
`Vis_history`	Compares simulated and historical data in plots.	off
`Vis_country`, `Vis_global17`, `Vis_map`	Selects country, regional, and map outputs.	on, on, off
`aggregate_global17`	Performs aggregation of results into 17-region, 10-region, and 5-region groupings.	on
`AgMIP_output`	Outputs results in AGMIP and IAMC_Template formats.	on
Scenario settings		
`socioeconomics`	Specifies the socioeconomic scenario. A file with the scenario name should exist in `setting/socioeconomic`.	SSP2
`climatepolicies`	Specifies the climate-policy scenario. A file with the scenario name should exist in `setting/climatepolicy`.	BaU
`climates`	Specifies the climate scenario. A file with the scenario name should exist in `setting/climate`.	NoCC
`runlist`	Allows direct scenario specification without using the scenario matrix. Scenario names are defined as `socioeconomic|climatepolicy|climate` (e.g., `SSP2|500C_CACN_DAC|NoCC`). If this item is specified, the three settings above are ignored.	–
`start_year`	Starting year of simulation.	2015
`target_year`	Ending year of simulation.	2100
Execution environment		
`run_parallel`	Runs multiple scenarios in parallel.	on
`serial_iterations`	Ordered list of solver iteration settings tried when `run_parallel` is `off`.	0, 3, 5
`MultiSol`	Specifies the number of solver candidates run concurrently for each year when `run_parallel` is `on`; the first successful solution is retained.	3
`ModelInt`	Optional suffix appended to scenario IDs and linked to `setting/modelint/<ModelInt>.gms`. If empty, no suffix is appended.	""
`NCPU`	Maximum number of CPUs used for parallel computation.	8
`year_timeout`	Maximum computation time allowed per simulation year (in seconds). If this time limit is exceeded, the scenario run is aborted.	900
`solprint`	Writes the full solve report into the listing files of `3_equilibrium.gms`, for debugging. Turning it off keeps the listing files small.	on

7.4 Execution Environment Requirements

The model can be executed under either Cygwin or Ubuntu with the following dependencies installed:

- **Cygwin** (required packages: `zip`, `unzip`)
- **Ubuntu**
- **R** (version ≥ 4.3.2; required libraries: `"tidyverse"`, `"gdxrrw"`, `"openxlsx"`, `"readxl"`, `"ggplot2"`, `"ggpmisc"`, `"RColorBrewer"`, `"purrr"`, `"furrr"`, `"scales"`, `"patchwork"`, `"sf"`, `"raster"`, `"ncdf4"`, `"exactextractr"`, `"rmapshaper"`, `"rnaturalearth"`,`"rnaturalearthdata"`, `"grDevices"`)
- **GAMS** (version ≥ 38, with the `PATH` solver available)
- **GitHub**

8 Development Guidelines

This section summarizes the conventions used when extending or maintaining AIM-ALPHA. The purpose is to keep the GAMS programs readable and to preserve consistency across calibration, equilibrium calculation, and output processing.

8.1 Location of Model Components

Sets and mappings shared by several programs are defined under `define`. Model assumptions and constant values are placed in the relevant file under `prog/inc_prog/const`. Their application conditions remain in the program that uses them. Source-data corrections are kept separate from economic equations and accompanied by a short explanation of the underlying data issue.

The model is processed in a sequence. Data are imported and regionalized in `1_data_import.gms` and `1-5_regression.gms`; base-year values are calibrated in `2_baseyear.gms`; and the dynamic equilibrium is solved in `3_equilibrium.gms`. Annual results are then combined and aggregated by the programs beginning with `4_` and `5_`. A change to a symbol or equation should therefore be followed through each affected stage, including the shell and R programs that read the resulting GDX files.

Numerical initialization, bounds, fixing, and scaling are maintained in `3_level.gms`, `3_lower.gms`, `3_fix.gms`, and `3_scale.gms`. Economic assumptions should not be hidden in these numerical-support files. The global `base_year` setting is used for calibration-year references instead of a year written directly into the program.

8.2 Symbol Names

Names are kept short enough for equations to remain readable, while retaining the economic meaning of each symbol. GAMS is case-insensitive, so capitalization is a convention for readers rather than a means of distinguishing symbols.

Table 13: General naming conventions

Symbol type	Convention	Examples
Variable	Begins with an uppercase letter	`QD`, `QSBas`, `PL`
Parameter	Begins with a lowercase letter	`qd0`, `qsBasPrev1`, `elaP`
Equation	Begins with an uppercase letter and ends in `Eq`	`QDEq`, `CroZeroProfitEq`
Set	Lowercase	`c`, `cty`, `a_liv`

Quantity variables begin with `Q`. The `QD` family contains total domestic demand and its components, such as `QDFood`, `QDFed`, and `QDInt`. Domestic production uses `QS`; production by basin, crop technology, and livestock system uses `QSBas`, `QSWat`, and `QSSys`, respectively. Land quantities use `QL`, and land prices use `PL`. Other common stems are `P` for prices, `S` for shares, `R` for rates, `Y` for yields, `C` for unit costs, and `N` for stocks or counts.

Short qualifiers are added in CamelCase when a distinction is needed. Examples include `Bas` for basin, `Cty` for country, `Cro` for crops, `Liv` for livestock, `Wat` for crop production technology, `Pas` for pasture, and `Emis` for emissions. A one-letter qualifier is avoided when it may have more than one interpretation.

Complementarity multipliers begin with `Mu` and name the associated constraint. Thus, `MuBasMin` is paired with `QSBasMinEq`, and `MuAfrCap` is paired with `AfrCapEq`. Equation names describe their role using terms such as `Bal`, `Logit`, `Min`, `Max`, `Parity`, and `ZeroProfit`.

8.2.1 Sets and Mappings

Basic Set names are lowercase and normally contain no more than three characters. A subset is named as the parent Set followed by its classification, as in `a_cro`, `a_liv_rum`, and `c_ntrd`. A two-dimensional mapping is named `<left>_<right>_map`; for example, `cty_reg_map(cty,reg)` maps countries to regions. Aliases use the same stem followed by `2`, as in `a_cro2`.

8.2.2 Time and Processing State

The same stem is retained as a value moves through calibration and the annual calculation. The suffixes `Prev1` and `Prev2` identify stored states one and two annual steps earlier. A bare `Prev` denotes the one-period-lagged working value for the current substep; when both stored states are needed, it is normally interpolated from `Prev2` and `Prev1`. `Ref` is reserved for a non-temporal benchmark or reference value, and `Drv` denotes an exogenous driver before the endogenous response is applied.

The suffix `_load` is reserved for data read from an external file before scenario or year selection. `Path` denotes a future path constructed inside the model, and `Raw` denotes a value before a bound or adjustment is applied. Bounds use the functional suffixes `Cap`, `Min`, and `Max`.

8.2.3 External Data and Output

Names fixed by an external GDX schema are not changed at the interface. When necessary, the symbol is renamed on the model side of the `$load` statement. Established abbreviations in imported databases, including the GTAP parameters `mmj`, `mmm`, `mme`, `tm`, and `te`, are retained.

Annual GDX outputs use `<stem>_result`. Combined time-series values use `<stem>_load`, single-year scratch values use `<stem>Single`, and aggregated values use `<stem>Agg_load`. A value derived from an endogenous variable retains the uppercase variable stem; an output derived from a Parameter retains the lowercase Parameter stem.

Descriptions are written as short noun phrases and include units where applicable. Standard forms include `[1000 t]`, `[1000 ha]`, `[1000 head]`, `[kt CO2-eq/yr]`, `[USD/t]`, and `[USD/ha]`.

8.3 Time Steps and Scenario Measures

The annual calculation can divide the transition from `pre_year` to `year` into smaller intervals. The loop in `3_equilibrium.gms` evaluates `3_timestep.gms`, applies the scenario measures in `3_scenario.gms`, and then solves the equilibrium. The solution levels from one pass provide the starting point for the next pass, which allows the model to approach the target year gradually when a direct annual change would be numerically difficult.

When `iter_num=0`, the loop is evaluated once and the requested year is solved directly. When `iter_num=k>0`, the loop runs from `n=0` to `n=k`. The first solve represents `pre_year`, the last represents `year`, and the intervening solves divide the transition into `k` equal intervals. There are therefore `k+1` equilibrium solves. The scalar `tCur` gives the time represented by the current pass, and the interpolation weights are

$$wPrev = \frac{k-n}{k}, \qquad wCur = \frac{n}{k}, \qquad tCur = pre_year + (year - pre_year)wCur. \tag{8.1}$$

Parameters that vary exogenously over time must be assigned on every pass through `3_timestep.gms`. A parameter stored at the two endpoint years is normally calculated as

$$x = x(pre_year)wPrev + x(year)wCur, \tag{8.2}$$

while a trend may instead be evaluated directly at `tCur`. This applies to drivers such as population, GDP per capita, elasticities, yield changes, carbon prices, and mitigation parameters. Quantities carried forward from earlier endogenous solutions use `Prev1` and `Prev2` for the stored annual states. Their one-period-lagged value at the current substep is

$$xPrev = xPrev2\, wPrev + xPrev1\, wCur. \tag{8.3}$$

At the final substep, and whenever `iter_num=0`, `xPrev` therefore equals `xPrev1`. Intermediate solves are used for numerical continuation and are not written as separate annual results; `3_result.gms` is evaluated only after the loop has finished.

An adjusted working driver must not overwrite its stored annual state. For example, `qsPrev1` and `qsBasPrev1` retain the exact previous-year solutions used to initialize `QS.L` and `QSBas.L`. The production floor is applied instead to `qsDrvPrev1` and `qsBasDrvPrev1`, which are used with their `Prev2` counterparts to construct the substep values `qsPrev` and `qsBasPrev`.

The order of calculation is important. `3_timestep.gms` first restores or constructs the baseline value for the current time point, after which `3_scenario.gms` applies the active measures. If a scenario changes a

Parameter in place, that Parameter must be restored or rebuilt on the next pass before the measure is applied again. Otherwise the policy effect compounds across substeps. Conversely, an exogenous Parameter that is set only outside the loop remains stale while the model advances toward the target year.

Introducing a time-varying exogenous driver therefore requires four related changes. Its annual series is declared and loaded outside the loop; its working Parameter is assigned in `3_timestep.gms` using `wPrev` and `wCur`, or directly from `tCur`; any fallback and backcasting behavior is specified there; and its baseline value is restored there if a scenario later modifies it in place. The implementation is checked at both endpoints: `n=0` must reproduce the `pre_year` value, `n=iter_num` must reproduce the target-year value, and `iter_num=0` must assign the target-year value directly.

A new scenario measure is added as an include file under `prog/inc_prog/scenario`. Its `SCE_*` switch is set to `off` by default, dispatched from `3_scenario.gms`, and enabled only in the relevant setting file. Symbols are declared outside the scenario include because that include is evaluated repeatedly.

Conditional compilation blocks use matching labels, for example `$ifthen.name` and `$endif.name`. Conditions over large domains include a sparse operand so that inactive tuples are not evaluated unnecessarily.

8.4 Validation of Changes

All affected GAMS stages are compiled with the settings used by the intended run. Changes to imported or exported symbols are tested through data import, base-year calibration, equilibrium calculation, and output aggregation. Modified shell and R programs are also parsed with their native tools.

For a rename or another change that is not intended to affect model behavior, a pre-change GDX is retained. Old and new symbols are mapped explicitly and their records are compared; both the domain and the values should agree exactly. A representative scenario run is then checked for GAMS execution errors, model status, solver status, and diagnostic residuals. A process exit code of zero alone does not establish that all annual solves succeeded.

Generated outputs, temporary solver files, and local configuration files are not included in a contribution unless they are an intentional part of the change.

9 Update History

2025-11-02 | R. Totake v1.0.0

Initial release of AIM-ALPHA documentation.

2026-2-11 | R. Totake v1.0.1

Added protected areas; split forests into primary and secondary categories; and divided pastureland into grazed and ungrazed categories to more accurately represent land use and grazing intensity.

2026-2-21 | R. Totake v1.0.2

Added an optional SSP-based urban land projection extension using the SEDAC 1-km urban dataset, while keeping the LUH2-consistent base-year urban area as the default.

2026-2-27 | R. Totake v1.0.3

Updated scenario and inter-model integration settings: introduced split scenario controls (`socioeconomic_setting`, `climatepolicy_setting`, `climate_setting`), standardized scenario suffix control with `ModelInt`, changed `runlist` format to `socioeconomic|climatepolicy|climate`, and added optional pre-run bridge processes (`hubauto`/Hub2ALPHA and `afoluauto`/AFOLU2ALPHA).

2026-08-29 | R. Totake v1.0.4

Updated the land-use and afforestation formulation, protected-area constraints, crop-yield drivers, inter-model integration, and model settings.

2026-10-18 | R. Totake v1.0.5

Standardized model symbol names and documentation, added development guidelines, and updated the regression scripts and figures. Updated price, subsidy and tax formulations, crop yields and production costs, forest and afforestation constraints, protected areas, price-data gap filling, and land-data processing.

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Appendix A: Production Units

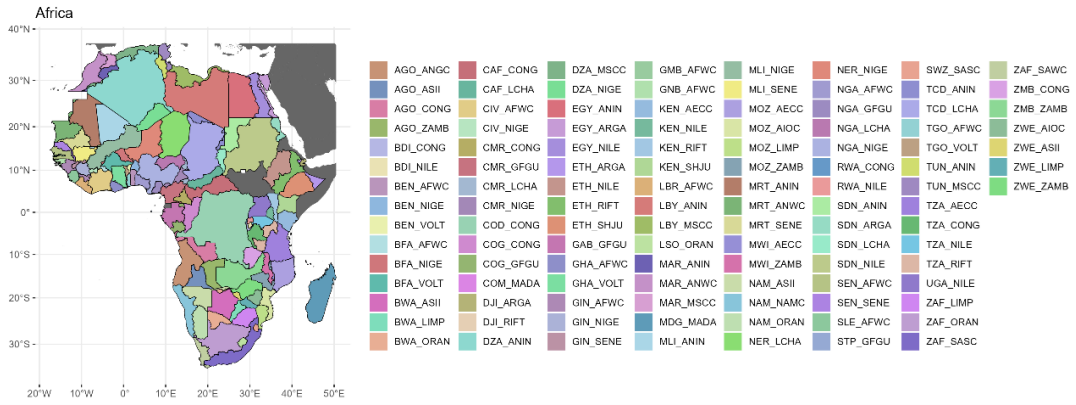


Figure A1: Production units in Africa

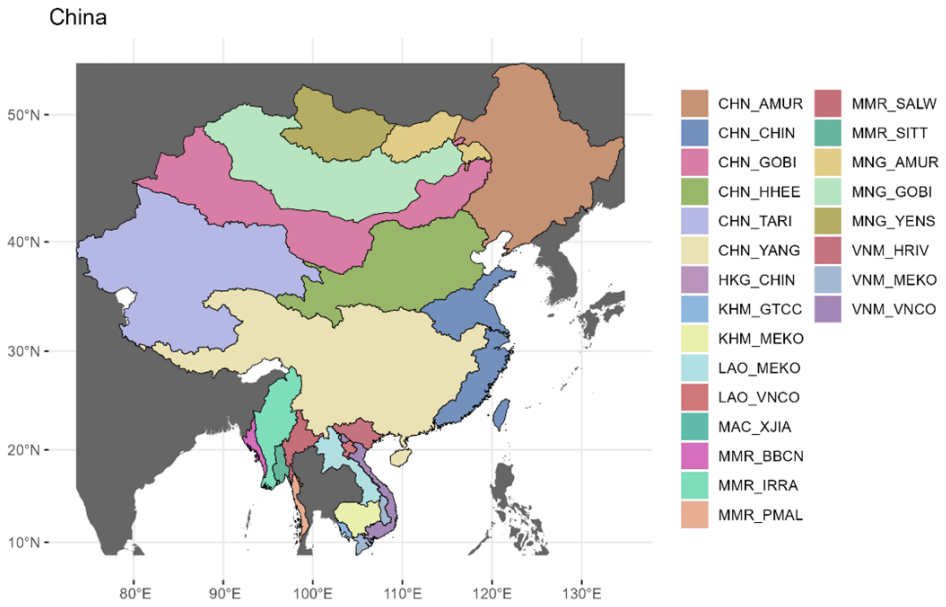


Figure A2: Production units in the China region

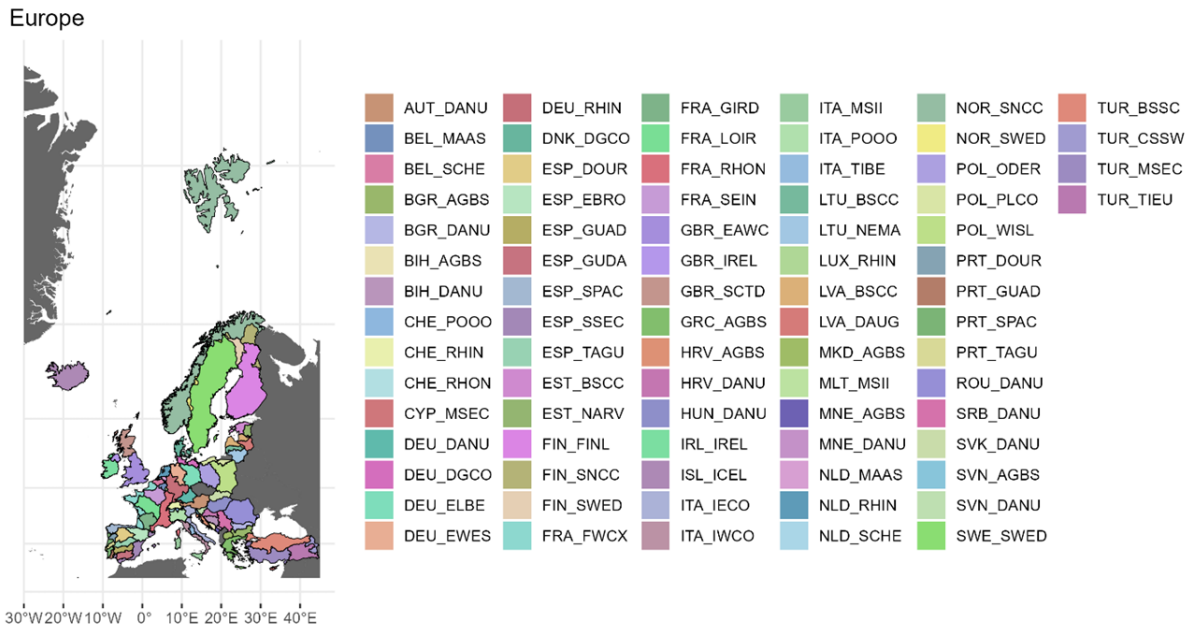


Figure A3: Production units in Europe

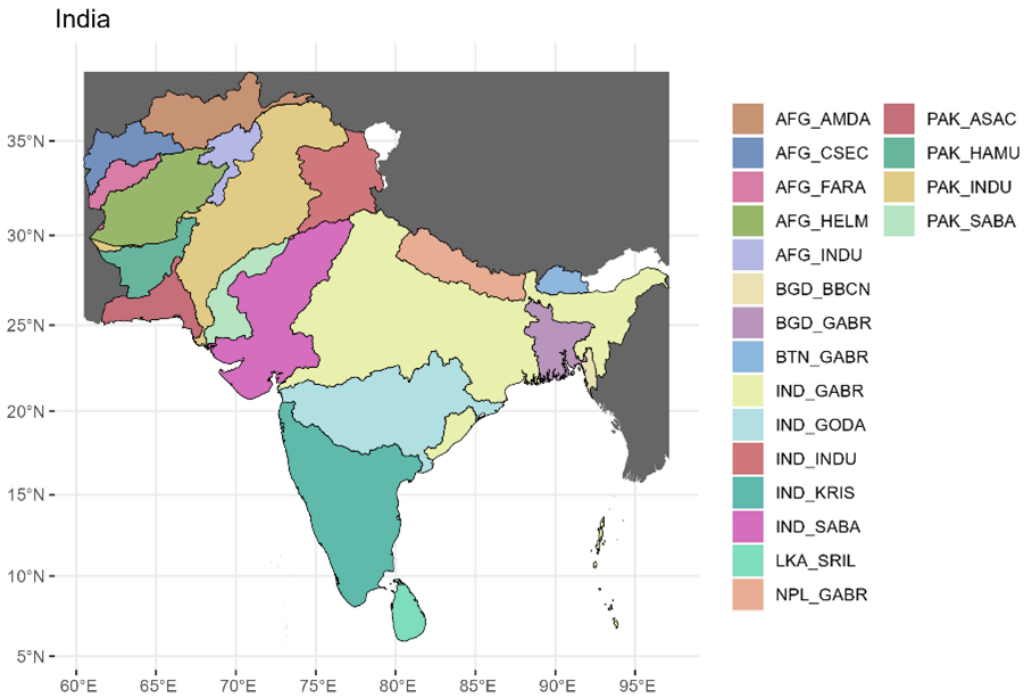


Figure A4: Production units in the India region

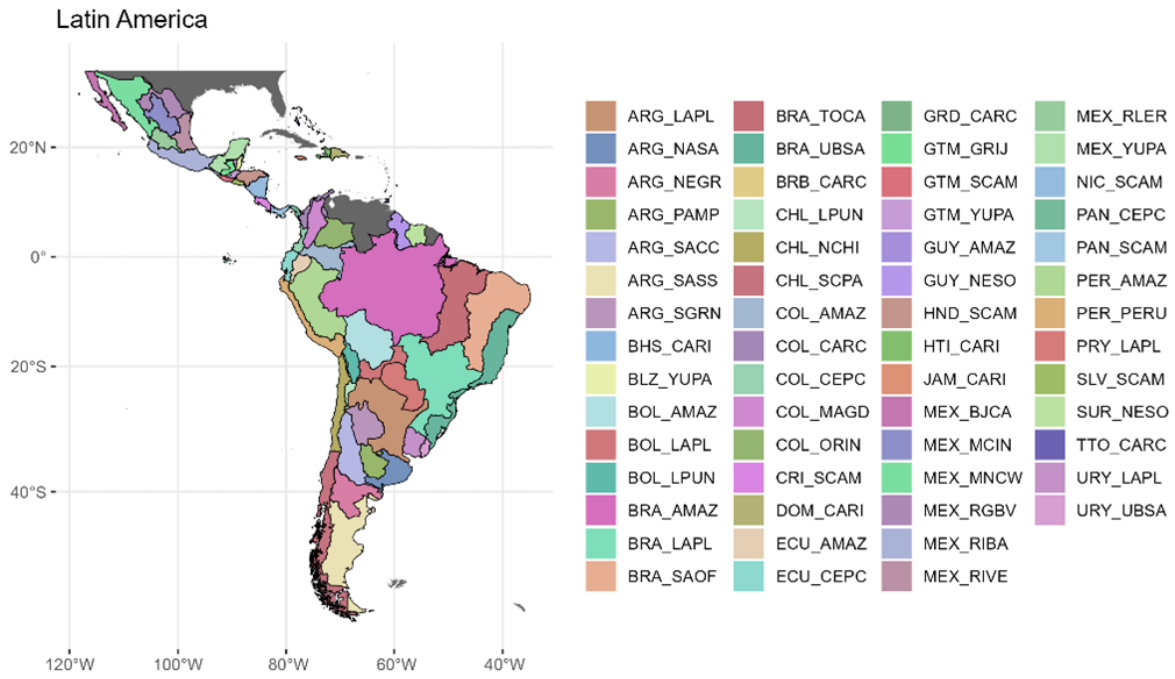


Figure A5: Production units in Latin America

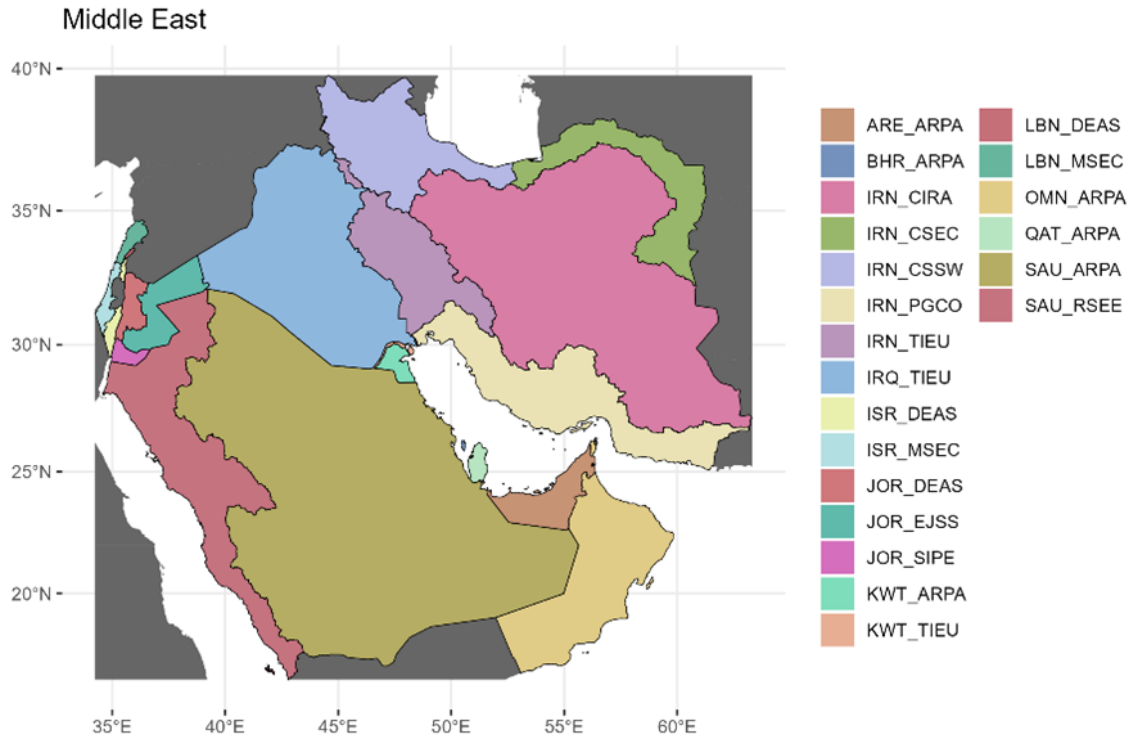


Figure A6: Production units in the Middle East

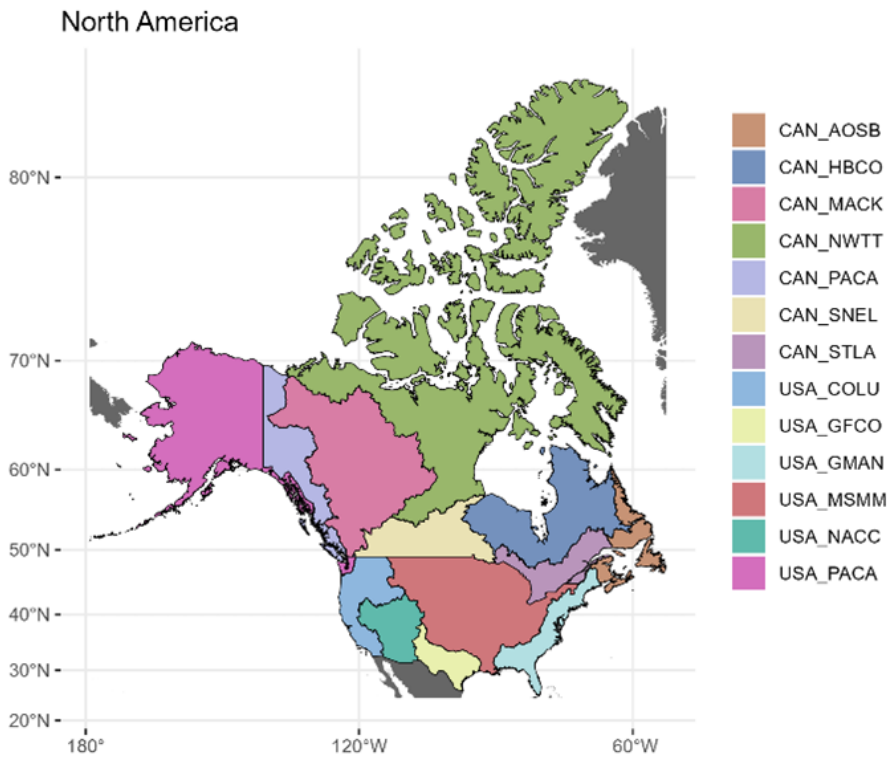


Figure A7: Production units in North America

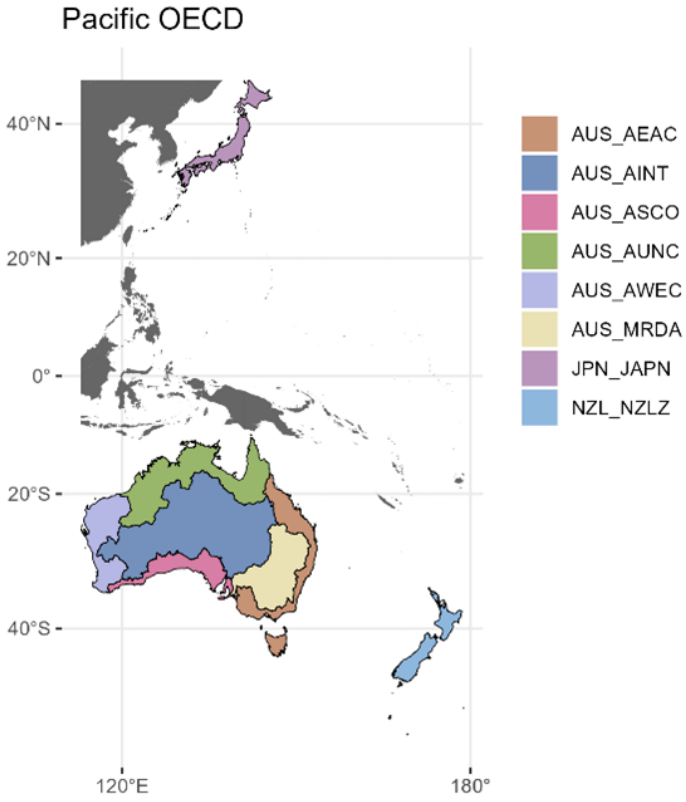


Figure A8: Production units in the Pacific OECD region

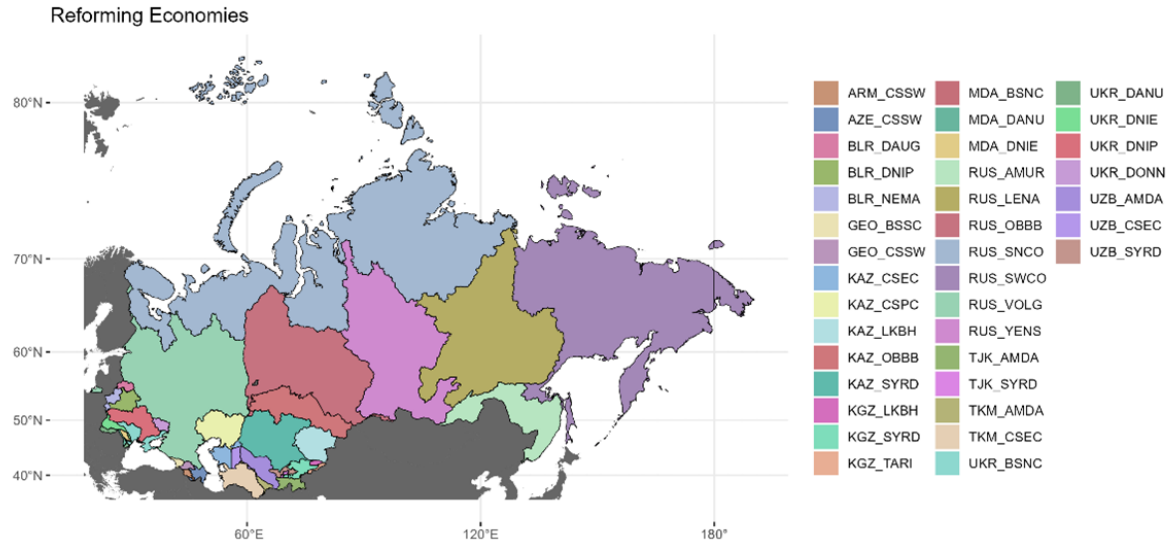


Figure A9: Production units in the former Soviet Union

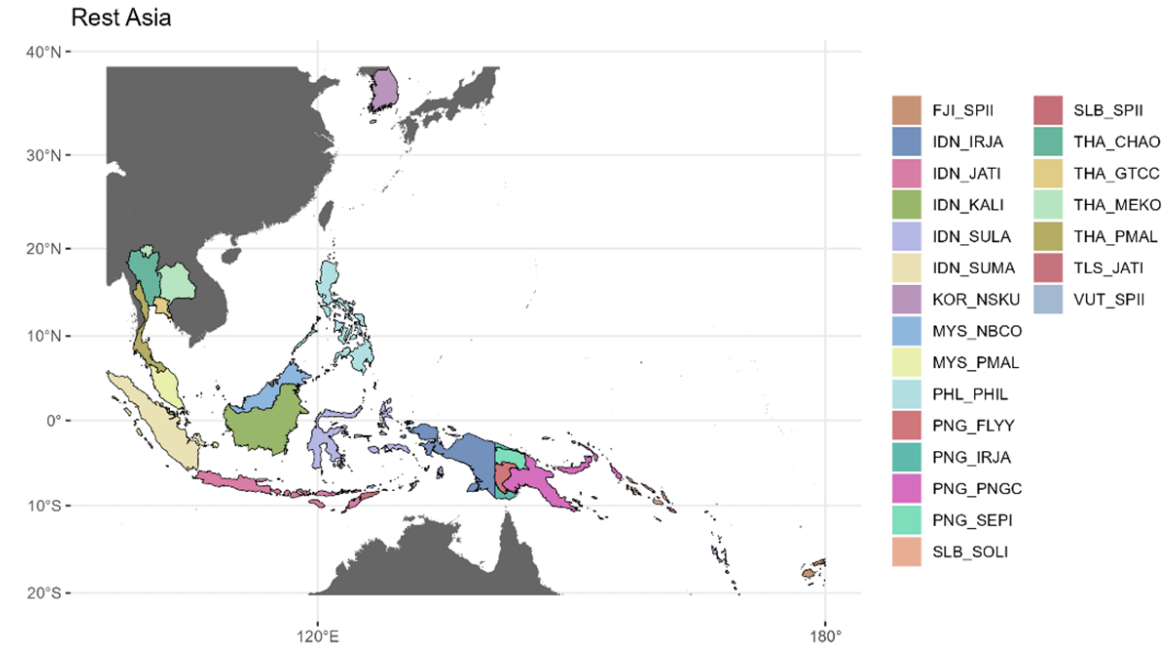


Figure A10: Production units in other Asian regions

Appendix B: Regional Aggregation

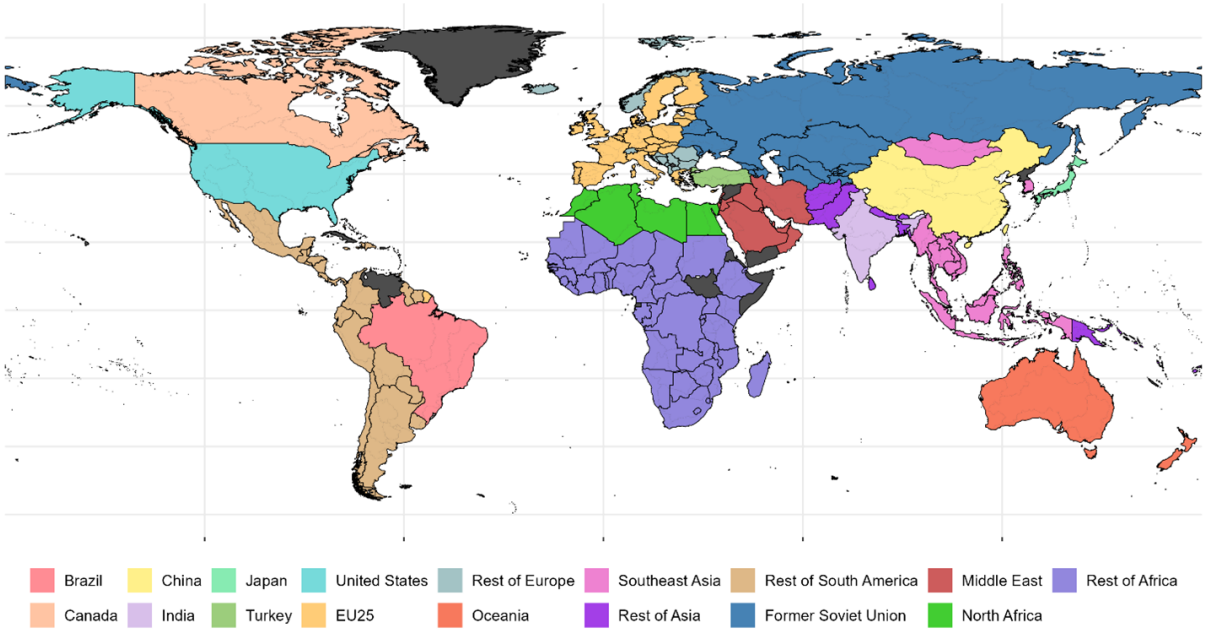


Figure B1: Correspondence between model countries/regions and 17 aggregated regions

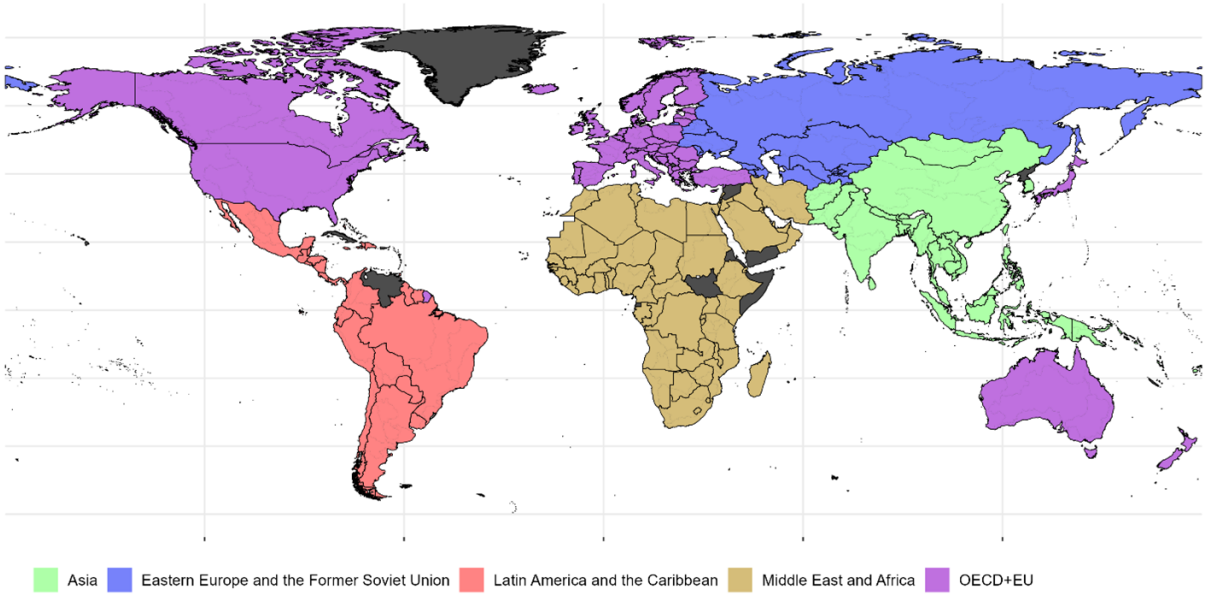


Figure B2: Correspondence between model countries/regions and R5 regions

Appendix C: Income Elasticities

Table C1 summarizes the projected income elasticities for 16 representative countries, selected to cover a wide range of regions and income levels.

Table C1: Projected income elasticities of 16 representative countries by scenario and year

Country	Scenario	Year	wht	rce	crl, mze, str	sgr	swt, stm, alc	pls, nut, ocr, spc	vol	vgt	frt	cmt, rmt, omt	pmt	omt	mlk, dai, egg
AUS	SSP1	2015	0.141	-0.081	-0.181	0.137	0.075	0.036	0.104	0.463	0.342	0.033	0.585	0.033	0.106
AUS	SSP1	2050	0.135	-0.092	-0.187	0.19	0.074	0.036	0.101	0.406	0.31	-0.005	0.55	-0.005	0.102
AUS	SSP1	2100	0.129	-0.103	-0.192	0.165	0.072	0.035	0.098	0.363	0.285	-0.043	0.515	-0.043	0.099
AUS	SSP2	2015	0.141	-0.081	-0.181	0.137	0.075	0.036	0.104	0.463	0.342	0.033	0.585	0.033	0.106
AUS	SSP2	2050	0.136	-0.09	-0.186	0.132	0.074	0.036	0.101	0.413	0.314	0.001	0.555	0.001	0.103
AUS	SSP2	2100	0.13	-0.1	-0.191	0.114	0.072	0.036	0.099	0.373	0.291	-0.033	0.524	-0.033	0.1
AUS	SSP3	2015	0.141	-0.081	-0.181	0.137	0.075	0.036	0.104	0.463	0.342	0.033	0.585	0.033	0.106
AUS	SSP3	2050	0.137	-0.088	-0.185	0.115	0.074	0.036	0.102	0.426	0.321	0.01	0.563	0.01	0.103
AUS	SSP3	2100	0.136	-0.089	-0.186	0.188	0.074	0.036	0.102	0.418	0.317	0.005	0.559	0.005	0.103
AUS	SSP4	2015	0.141	-0.081	-0.181	0.137	0.075	0.036	0.104	0.463	0.342	0.033	0.585	0.033	0.106
AUS	SSP4	2050	0.135	-0.092	-0.187	0.198	0.074	0.036	0.101	0.405	0.31	-0.005	0.55	-0.005	0.102
AUS	SSP4	2100	0.13	-0.101	-0.191	0.114	0.072	0.036	0.098	0.37	0.289	-0.037	0.521	-0.037	0.1
AUS	SSP5	2015	0.141	-0.081	-0.181	0.137	0.075	0.036	0.104	0.463	0.342	0.033	0.585	0.033	0.106
AUS	SSP5	2050	0.134	-0.093	-0.188	0.131	0.073	0.036	0.1	0.4	0.307	-0.009	0.546	-0.009	0.102
AUS	SSP5	2100	0.125	-0.111	-0.196	0.114	0.071	0.035	0.096	0.342	0.271	-0.067	0.493	-0.067	0.097
BRA	SSP1	2015	0.189	0.079	-0.112	0.088	0.088	0.015	0.141	0.755	0.365	0.098	0.456	0.098	0.142
BRA	SSP1	2050	0.177	0.07	-0.113	0.085	0.085	0.015	0.133	0.583	0.319	0.041	0.443	0.041	0.134
BRA	SSP1	2100	0.16	0.06	-0.113	0.08	0.08	0.015	0.124	0.451	0.273	-0.055	0.42	-0.055	0.124
BRA	SSP2	2015	0.189	0.079	-0.112	0.088	0.088	0.015	0.141	0.755	0.365	0.098	0.456	0.098	0.142
BRA	SSP2	2050	0.178	0.071	-0.113	0.085	0.085	0.015	0.134	0.598	0.323	0.048	0.445	0.048	0.135
BRA	SSP2	2100	0.161	0.06	-0.113	0.081	0.081	0.015	0.124	0.459	0.276	-0.047	0.422	-0.047	0.125
BRA	SSP3	2015	0.189	0.079	-0.112	0.088	0.088	0.015	0.141	0.755	0.365	0.098	0.456	0.098	0.142
BRA	SSP3	2050	0.18	0.073	-0.113	0.085	0.085	0.015	0.135	0.623	0.331	0.058	0.447	0.058	0.136
BRA	SSP3	2100	0.171	0.067	-0.114	0.083	0.083	0.015	0.13	0.532	0.303	0.013	0.437	0.013	0.131
BRA	SSP4	2015	0.189	0.079	-0.112	0.088	0.088	0.015	0.141	0.755	0.365	0.098	0.456	0.098	0.142
BRA	SSP4	2050	0.178	0.071	-0.113	0.085	0.085	0.015	0.134	0.596	0.323	0.047	0.444	0.047	0.135
BRA	SSP4	2100	0.161	0.06	-0.113	0.081	0.081	0.015	0.124	0.455	0.275	-0.051	0.421	-0.051	0.125
BRA	SSP5	2015	0.189	0.079	-0.112	0.088	0.088	0.015	0.141	0.755	0.365	0.098	0.456	0.098	0.142
BRA	SSP5	2050	0.175	0.069	-0.113	0.084	0.084	0.015	0.132	0.563	0.313	0.031	0.441	0.031	0.133
BRA	SSP5	2100	0.155	0.056	-0.113	0.079	0.079	0.015	0.12	0.419	0.261	-0.097	0.409	-0.097	0.121
CHN	SSP1	2015	0.164	0.05	-0.348	0.252	0.252	0.033	0.344	0.199	0.144	0.108	0.913	0.108	0.288
CHN	SSP1	2050	0.152	0.044	-0.406	0.225	0.225	0.033	0.296	0.181	0.136	0.063	0.792	0.063	0.253
CHN	SSP1	2100	0.143	0.04	-0.476	0.209	0.209	0.032	0.269	0.17	0.13	0.029	0.719	0.029	0.233
CHN	SSP2	2015	0.164	0.05	-0.348	0.252	0.252	0.033	0.344	0.199	0.144	0.108	0.913	0.108	0.288
CHN	SSP2	2050	0.153	0.044	-0.4	0.227	0.227	0.033	0.3	0.183	0.136	0.067	0.801	0.067	0.256
CHN	SSP2	2100	0.144	0.04	-0.466	0.21	0.21	0.032	0.272	0.171	0.131	0.033	0.727	0.033	0.235
CHN	SSP3	2015	0.164	0.05	-0.348	0.252	0.252	0.033	0.344	0.199	0.144	0.108	0.913	0.108	0.288
CHN	SSP3	2050	0.155	0.045	-0.389	0.231	0.231	0.033	0.306	0.185	0.138	0.074	0.818	0.074	0.261
CHN	SSP3	2100	0.151	0.044	-0.408	0.224	0.224	0.033	0.295	0.181	0.135	0.062	0.788	0.062	0.252
CHN	SSP4	2015	0.164	0.05	-0.348	0.252	0.252	0.033	0.344	0.199	0.144	0.108	0.913	0.108	0.288
CHN	SSP4	2050	0.152	0.044	-0.403	0.226	0.226	0.033	0.298	0.182	0.136	0.065	0.796	0.065	0.255
CHN	SSP4	2100	0.144	0.04	-0.467	0.21	0.21	0.032	0.272	0.171	0.131	0.032	0.726	0.032	0.235
CHN	SSP5	2015	0.164	0.05	-0.348	0.252	0.252	0.033	0.344	0.199	0.144	0.108	0.913	0.108	0.288
CHN	SSP5	2050	0.15	0.043	-0.416	0.222	0.222	0.033	0.291	0.179	0.135	0.057	0.778	0.057	0.249
CHN	SSP5	2100	0.138	0.038	-0.539	0.2	0.2	0.032	0.256	0.165	0.128	0.009	0.682	0.009	0.223
EGY	SSP1	2015	0.147	-0.008	-0.185	0.14	0.14	0.027	0.438	0.211	0.164	0.523	0.866	0.523	0.381
EGY	SSP1	2050	0.133	-0.021	-0.198	0.129	0.129	0.027	0.348	0.187	0.15	0.318	0.717	0.318	0.31
EGY	SSP1	2100	0.12	-0.032	-0.209	0.12	0.12	0.026	0.294	0.169	0.139	0.177	0.631	0.177	0.266
EGY	SSP2	2015	0.147	-0.008	-0.185	0.14	0.14	0.027	0.438	0.211	0.164	0.523	0.866	0.523	0.381
EGY	SSP2	2050	0.135	-0.019	-0.196	0.13	0.13	0.027	0.36	0.191	0.152	0.346	0.736	0.346	0.32
EGY	SSP2	2100	0.123	-0.03	-0.207	0.121	0.121	0.027	0.303	0.172	0.141	0.203	0.646	0.203	0.274
EGY	SSP3	2015	0.147	-0.008	-0.185	0.14	0.14	0.027	0.438	0.211	0.164	0.523	0.866	0.523	0.381
EGY	SSP3	2050	0.139	-0.016	-0.193	0.133	0.133	0.027	0.378	0.196	0.156	0.388	0.765	0.388	0.335
EGY	SSP3	2100	0.134	-0.02	-0.197	0.129	0.129	0.027	0.352	0.188	0.151	0.329	0.724	0.329	0.314
EGY	SSP4	2015	0.147	-0.008	-0.185	0.14	0.14	0.027	0.438	0.211	0.164	0.523	0.866	0.523	0.381
EGY	SSP4	2050	0.137	-0.017	-0.194	0.132	0.132	0.027	0.371	0.194	0.154	0.373	0.754	0.373	0.329
EGY	SSP4	2100	0.127	-0.026	-0.202	0.125	0.125	0.027	0.323	0.179	0.145	0.257	0.678	0.257	0.29
EGY	SSP5	2015	0.147	-0.008	-0.185	0.14	0.14	0.027	0.438	0.211	0.164	0.523	0.866	0.523	0.381
EGY	SSP5	2050	0.131	-0.023	-0.199	0.127	0.127	0.027	0.339	0.184	0.149	0.297	0.704	0.297	0.303
EGY	SSP5	2100	0.116	-0.036	-0.213	0.117	0.117	0.026	0.279	0.163	0.136	0.129	0.604	0.129	0.254
DEU	SSP1	2015	0.139	-0.383	-0.119	0.118	0.06	0.033	0.116	0.406	0.322	0.163	1.036	0.163	0.099
DEU	SSP1	2050	0.135	-0.435	-0.119	0.091	0.059	0.033	0.113	0.375	0.302	0.134	0.947	0.134	0.097
DEU	SSP1	2100	0.13	-0.511	-0.119	0.078	0.058	0.033	0.11	0.347	0.283	0.104	0.868	0.104	0.095

Table C1: Projected income elasticities by country, scenario, and year (continued)

Country	Scenario	Year	wht	rce	crl, mze, str	sgr	swt, stm, alc	pls, nut, ocr, spc	vol	vgt	frt	cmt, rmt, omt	pmt	omt	mlk, dai, egg
DEU	SSP2	2015	0.139	-0.383	-0.119	0.118	0.06	0.033	0.116	0.406	0.322	0.163	1.036	0.163	0.099
DEU	SSP2	2050	0.135	-0.427	-0.119	0.091	0.059	0.033	0.113	0.379	0.304	0.138	0.958	0.138	0.097
DEU	SSP2	2100	0.131	-0.491	-0.119	0.103	0.059	0.033	0.111	0.353	0.287	0.11	0.884	0.11	0.095
DEU	SSP3	2015	0.139	-0.383	-0.119	0.118	0.06	0.033	0.116	0.406	0.322	0.163	1.036	0.163	0.099
DEU	SSP3	2050	0.136	-0.411	-0.119	0.1	0.06	0.033	0.114	0.387	0.31	0.146	0.982	0.146	0.098
DEU	SSP3	2100	0.137	-0.405	-0.119	0.099	0.06	0.033	0.114	0.391	0.312	0.149	0.993	0.149	0.098
DEU	SSP4	2015	0.139	-0.383	-0.119	0.118	0.06	0.033	0.116	0.406	0.322	0.163	1.036	0.163	0.099
DEU	SSP4	2050	0.135	-0.435	-0.119	0.091	0.059	0.033	0.113	0.375	0.302	0.133	0.946	0.133	0.097
DEU	SSP4	2100	0.131	-0.495	-0.119	0.079	0.059	0.033	0.11	0.351	0.287	0.109	0.881	0.109	0.095
DEU	SSP5	2015	0.139	-0.383	-0.119	0.118	0.06	0.033	0.116	0.406	0.322	0.163	1.036	0.163	0.099
DEU	SSP5	2050	0.134	-0.447	-0.119	0.09	0.059	0.033	0.112	0.369	0.298	0.128	0.931	0.128	0.096
DEU	SSP5	2100	0.126	-0.616	-0.118	0.091	0.058	0.033	0.108	0.325	0.269	0.079	0.807	0.079	0.093
IND	SSP1	2015	0.18	0.058	-0.325	0.177	0.177	0.014	0.254	0.659	0.25	3.346	3.655	3.346	0.25
IND	SSP1	2050	0.163	0.049	-0.412	0.159	0.159	0.014	0.218	0.472	0.219	1.214	1.583	1.214	0.215
IND	SSP1	2100	0.15	0.043	-0.564	0.147	0.147	0.014	0.197	0.391	0.201	0.785	1.203	0.785	0.194
IND	SSP2	2015	0.18	0.058	-0.325	0.177	0.177	0.014	0.254	0.659	0.25	3.346	3.655	3.346	0.25
IND	SSP2	2050	0.165	0.05	-0.4	0.16	0.16	0.014	0.221	0.485	0.221	1.299	1.661	1.299	0.218
IND	SSP2	2100	0.152	0.044	-0.53	0.148	0.148	0.014	0.2	0.401	0.203	0.833	1.244	0.833	0.197
IND	SSP3	2015	0.18	0.058	-0.325	0.177	0.177	0.014	0.254	0.659	0.25	3.346	3.655	3.346	0.25
IND	SSP3	2050	0.167	0.051	-0.383	0.163	0.163	0.014	0.227	0.508	0.226	1.455	1.807	1.455	0.223
IND	SSP3	2100	0.162	0.049	-0.418	0.158	0.158	0.014	0.216	0.465	0.217	1.174	1.546	1.174	0.213
IND	SSP4	2015	0.18	0.058	-0.325	0.177	0.177	0.014	0.254	0.659	0.25	3.346	3.655	3.346	0.25
IND	SSP4	2050	0.166	0.051	-0.389	0.162	0.162	0.014	0.225	0.499	0.224	1.39	1.746	1.39	0.221
IND	SSP4	2100	0.156	0.046	-0.471	0.152	0.152	0.014	0.207	0.426	0.209	0.96	1.354	0.96	0.204
IND	SSP5	2015	0.18	0.058	-0.325	0.177	0.177	0.014	0.254	0.659	0.25	3.346	3.655	3.346	0.25
IND	SSP5	2050	0.161	0.048	-0.425	0.157	0.157	0.014	0.215	0.458	0.216	1.136	1.511	1.136	0.212
IND	SSP5	2100	0.145	0.04	-0.713	0.142	0.142	0.014	0.189	0.367	0.195	0.67	1.108	0.67	0.187
IDN	SSP1	2015	0.316	0.069	-0.264	0.136	0.219	0.032	0.36	0.89	0.257	0.75	1.167	0.75	0.921
IDN	SSP1	2050	0.28	0.063	-0.295	0.083	0.197	0.031	0.306	0.635	0.233	0.498	0.946	0.498	0.652
IDN	SSP1	2100	0.256	0.057	-0.326	0.181	0.184	0.031	0.275	0.528	0.218	0.374	0.845	0.374	0.54
IDN	SSP2	2015	0.316	0.069	-0.264	0.136	0.219	0.032	0.36	0.89	0.257	0.75	1.167	0.75	0.921
IDN	SSP2	2050	0.284	0.063	-0.291	0.083	0.2	0.031	0.312	0.657	0.235	0.521	0.966	0.521	0.674
IDN	SSP2	2100	0.26	0.058	-0.321	0.202	0.186	0.031	0.28	0.543	0.22	0.393	0.859	0.393	0.556
IDN	SSP3	2015	0.316	0.069	-0.264	0.136	0.219	0.032	0.36	0.89	0.257	0.75	1.167	0.75	0.921
IDN	SSP3	2050	0.291	0.065	-0.284	0.084	0.204	0.031	0.321	0.695	0.24	0.561	1	0.561	0.715
IDN	SSP3	2100	0.283	0.063	-0.292	0.113	0.199	0.031	0.31	0.649	0.234	0.512	0.958	0.512	0.666
IDN	SSP4	2015	0.316	0.069	-0.264	0.136	0.219	0.032	0.36	0.89	0.257	0.75	1.167	0.75	0.921
IDN	SSP4	2050	0.288	0.064	-0.287	0.082	0.202	0.031	0.317	0.679	0.238	0.544	0.986	0.544	0.698
IDN	SSP4	2100	0.269	0.06	-0.308	0.218	0.191	0.031	0.291	0.581	0.226	0.437	0.896	0.437	0.595
IDN	SSP5	2015	0.316	0.069	-0.264	0.136	0.219	0.032	0.36	0.89	0.257	0.75	1.167	0.75	0.921
IDN	SSP5	2050	0.276	0.062	-0.3	0.082	0.195	0.031	0.3	0.613	0.23	0.473	0.926	0.473	0.628
IDN	SSP5	2100	0.245	0.055	-0.348	0.19	0.177	0.031	0.262	0.487	0.211	0.32	0.801	0.32	0.497
JPN	SSP1	2015	0.159	0.054	-0.393	0.069	0.111	0.021	0.213	0.65	0.418	0.231	1.037	0.231	0.211
JPN	SSP1	2050	0.153	0.051	-0.426	0.07	0.108	0.021	0.202	0.56	0.38	0.184	0.938	0.184	0.201
JPN	SSP1	2100	0.146	0.048	-0.474	0.085	0.105	0.021	0.191	0.493	0.348	0.141	0.851	0.141	0.19
JPN	SSP2	2015	0.159	0.054	-0.393	0.069	0.111	0.021	0.213	0.65	0.418	0.231	1.037	0.231	0.211
JPN	SSP2	2050	0.154	0.052	-0.42	0.194	0.109	0.021	0.204	0.574	0.386	0.192	0.954	0.192	0.202
JPN	SSP2	2100	0.148	0.049	-0.46	0.083	0.106	0.021	0.194	0.508	0.356	0.152	0.872	0.152	0.193
JPN	SSP3	2015	0.159	0.054	-0.393	0.069	0.111	0.021	0.213	0.65	0.418	0.231	1.037	0.231	0.211
JPN	SSP3	2050	0.156	0.053	-0.409	0.2	0.11	0.021	0.207	0.6	0.397	0.206	0.984	0.206	0.206
JPN	SSP3	2100	0.156	0.053	-0.407	0.154	0.11	0.021	0.207	0.604	0.399	0.208	0.988	0.208	0.206
JPN	SSP4	2015	0.159	0.054	-0.393	0.069	0.111	0.021	0.213	0.65	0.418	0.231	1.037	0.231	0.211
JPN	SSP4	2050	0.153	0.051	-0.427	0.07	0.108	0.021	0.202	0.56	0.38	0.184	0.937	0.184	0.2
JPN	SSP4	2100	0.148	0.049	-0.464	0.083	0.106	0.021	0.193	0.504	0.354	0.149	0.866	0.149	0.192
JPN	SSP5	2015	0.159	0.054	-0.393	0.069	0.111	0.021	0.213	0.65	0.418	0.231	1.037	0.231	0.211
JPN	SSP5	2050	0.151	0.051	-0.436	0.077	0.108	0.021	0.199	0.543	0.372	0.174	0.916	0.174	0.198
JPN	SSP5	2100	0.142	0.046	-0.524	0.091	0.103	0.021	0.184	0.452	0.328	0.111	0.794	0.111	0.183
MEX	SSP1	2015	0.341	0.174	-0.042	0.086	0.086	0.024	0.247	0.577	0.44	0.219	0.64	0.219	0.172
MEX	SSP1	2050	0.3	0.133	-0.04	0.084	0.084	0.024	0.226	0.476	0.378	0.148	0.599	0.148	0.161
MEX	SSP1	2100	0.269	0.101	-0.038	0.081	0.081	0.024	0.209	0.411	0.335	0.084	0.563	0.084	0.152
MEX	SSP2	2015	0.341	0.174	-0.042	0.086	0.086	0.024	0.247	0.577	0.44	0.219	0.64	0.219	0.172
MEX	SSP2	2050	0.306	0.139	-0.04	0.084	0.084	0.024	0.229	0.489	0.387	0.159	0.605	0.159	0.163
MEX	SSP2	2100	0.274	0.106	-0.038	0.081	0.081	0.024	0.211	0.42	0.342	0.094	0.569	0.094	0.154

Table C1: Projected income elasticities by country, scenario, and year (continued)

Country	Scenario	Year	wht	rce	crl, mze, str	sgr	swt, stm, alc	pls, nut, ocr, spc	vol	vgt	frt	cmt, rmt, omt	pmt	omt	mlk, dai, egg
MEX	SSP3	2015	0.341	0.174	-0.042	0.086	0.086	0.024	0.247	0.577	0.44	0.219	0.64	0.219	0.172
MEX	SSP3	2050	0.317	0.149	-0.041	0.085	0.085	0.024	0.234	0.514	0.402	0.178	0.616	0.178	0.166
MEX	SSP3	2100	0.313	0.146	-0.041	0.084	0.084	0.024	0.232	0.505	0.396	0.171	0.612	0.171	0.165
MEX	SSP4	2015	0.341	0.174	-0.042	0.086	0.086	0.024	0.247	0.577	0.44	0.219	0.64	0.219	0.172
MEX	SSP4	2050	0.306	0.139	-0.04	0.084	0.084	0.024	0.229	0.489	0.386	0.159	0.605	0.159	0.163
MEX	SSP4	2100	0.273	0.106	-0.038	0.081	0.081	0.024	0.211	0.42	0.341	0.094	0.569	0.094	0.154
MEX	SSP5	2015	0.341	0.174	-0.042	0.086	0.086	0.024	0.247	0.577	0.44	0.219	0.64	0.219	0.172
MEX	SSP5	2050	0.293	0.126	-0.04	0.083	0.083	0.024	0.222	0.46	0.368	0.135	0.591	0.135	0.159
MEX	SSP5	2100	0.253	0.085	-0.037	0.08	0.08	0.024	0.2	0.381	0.315	0.045	0.542	0.045	0.148
NPL	SSP1	2015	0.228	0.079	-0.195	0.479	0.479	0.016	0.286	0.462	0.346	1.868	2.259	1.868	0.325
NPL	SSP1	2050	0.206	0.069	-0.224	0.369	0.369	0.015	0.241	0.358	0.284	0.903	1.336	0.903	0.269
NPL	SSP1	2100	0.182	0.057	-0.265	0.298	0.298	0.015	0.207	0.291	0.239	0.564	1.034	0.564	0.227
NPL	SSP2	2015	0.228	0.079	-0.195	0.479	0.479	0.016	0.286	0.462	0.346	1.868	2.259	1.868	0.325
NPL	SSP2	2050	0.209	0.07	-0.219	0.381	0.381	0.015	0.247	0.37	0.291	0.975	1.403	0.975	0.275
NPL	SSP2	2100	0.185	0.059	-0.259	0.305	0.305	0.015	0.21	0.298	0.244	0.594	1.06	0.594	0.231
NPL	SSP3	2015	0.228	0.079	-0.195	0.479	0.479	0.016	0.286	0.462	0.346	1.868	2.259	1.868	0.325
NPL	SSP3	2050	0.215	0.073	-0.212	0.405	0.405	0.015	0.257	0.393	0.305	1.128	1.545	1.128	0.288
NPL	SSP3	2100	0.203	0.067	-0.228	0.358	0.358	0.015	0.236	0.348	0.277	0.846	1.285	0.846	0.263
NPL	SSP4	2015	0.228	0.079	-0.195	0.479	0.479	0.016	0.286	0.462	0.346	1.868	2.259	1.868	0.325
NPL	SSP4	2050	0.212	0.072	-0.215	0.395	0.395	0.015	0.253	0.383	0.299	1.058	1.48	1.058	0.283
NPL	SSP4	2100	0.192	0.062	-0.246	0.324	0.324	0.015	0.22	0.315	0.256	0.677	1.133	0.677	0.243
NPL	SSP5	2015	0.228	0.079	-0.195	0.479	0.479	0.016	0.286	0.462	0.346	1.868	2.259	1.868	0.325
NPL	SSP5	2050	0.202	0.067	-0.229	0.355	0.355	0.015	0.235	0.346	0.276	0.831	1.271	0.831	0.261
NPL	SSP5	2100	0.176	0.054	-0.282	0.282	0.282	0.015	0.198	0.276	0.228	0.494	0.973	0.494	0.217
NGA	SSP1	2015	0.429	0.296	-0.064	0.247	0.247	0.017	0.286	0.873	0.362	1.985	2.945	1.985	1.875
NGA	SSP1	2050	0.38	0.252	-0.065	0.225	0.225	0.017	0.258	0.66	0.319	1.126	1.972	1.126	1.127
NGA	SSP1	2100	0.31	0.193	-0.064	0.194	0.194	0.017	0.218	0.469	0.263	0.635	1.369	0.635	0.699
NGA	SSP2	2015	0.429	0.296	-0.064	0.247	0.247	0.017	0.286	0.873	0.362	1.985	2.945	1.985	1.875
NGA	SSP2	2050	0.391	0.262	-0.065	0.23	0.23	0.017	0.264	0.7	0.328	1.252	2.121	1.252	1.239
NGA	SSP2	2100	0.325	0.205	-0.064	0.201	0.201	0.017	0.227	0.502	0.274	0.713	1.467	0.713	0.764
NGA	SSP3	2015	0.429	0.296	-0.064	0.247	0.247	0.017	0.286	0.873	0.362	1.985	2.945	1.985	1.875
NGA	SSP3	2050	0.404	0.274	-0.064	0.236	0.236	0.017	0.272	0.754	0.34	1.441	2.339	1.441	1.406
NGA	SSP3	2100	0.375	0.248	-0.065	0.223	0.223	0.017	0.255	0.645	0.315	1.082	1.919	1.082	1.087
NGA	SSP4	2015	0.429	0.296	-0.064	0.247	0.247	0.017	0.286	0.873	0.362	1.985	2.945	1.985	1.875
NGA	SSP4	2050	0.401	0.271	-0.065	0.235	0.235	0.017	0.27	0.739	0.337	1.384	2.274	1.384	1.356
NGA	SSP4	2100	0.349	0.225	-0.065	0.211	0.211	0.017	0.24	0.563	0.293	0.862	1.652	0.862	0.893
NGA	SSP5	2015	0.429	0.296	-0.064	0.247	0.247	0.017	0.286	0.873	0.362	1.985	2.945	1.985	1.875
NGA	SSP5	2050	0.372	0.245	-0.065	0.222	0.222	0.017	0.253	0.633	0.312	1.049	1.879	1.049	1.058
NGA	SSP5	2100	0.293	0.179	-0.063	0.186	0.186	0.017	0.209	0.434	0.25	0.552	1.265	0.552	0.631
RUS	SSP1	2015	0.127	-0.506	-0.243	0.222	0.08	0.092	0.167	0.502	0.367	0.192	0.698	0.192	0.129
RUS	SSP1	2050	0.122	-0.627	-0.254	0.177	0.079	0.09	0.16	0.441	0.334	0.146	0.662	0.146	0.124
RUS	SSP1	2100	0.114	-1.132	-0.273	0.085	0.076	0.086	0.15	0.377	0.296	0.081	0.61	0.081	0.118
RUS	SSP2	2015	0.127	-0.506	-0.243	0.222	0.08	0.092	0.167	0.502	0.367	0.192	0.698	0.192	0.129
RUS	SSP2	2050	0.122	-0.613	-0.253	0.194	0.079	0.09	0.16	0.446	0.336	0.15	0.665	0.15	0.124
RUS	SSP2	2100	0.115	-1.002	-0.271	0.083	0.076	0.087	0.151	0.384	0.3	0.089	0.616	0.089	0.119
RUS	SSP3	2015	0.127	-0.506	-0.243	0.222	0.08	0.092	0.167	0.502	0.367	0.192	0.698	0.192	0.129
RUS	SSP3	2050	0.123	-0.592	-0.252	0.128	0.079	0.09	0.161	0.454	0.341	0.157	0.67	0.157	0.125
RUS	SSP3	2100	0.12	-0.684	-0.258	0.087	0.078	0.089	0.157	0.425	0.324	0.132	0.65	0.132	0.123
RUS	SSP4	2015	0.127	-0.506	-0.243	0.222	0.08	0.092	0.167	0.502	0.367	0.192	0.698	0.192	0.129
RUS	SSP4	2050	0.122	-0.627	-0.254	0.198	0.079	0.09	0.16	0.441	0.334	0.146	0.662	0.146	0.124
RUS	SSP4	2100	0.115	-1.038	-0.271	0.083	0.076	0.087	0.15	0.382	0.299	0.087	0.614	0.087	0.118
RUS	SSP5	2015	0.127	-0.506	-0.243	0.222	0.08	0.092	0.167	0.502	0.367	0.192	0.698	0.192	0.129
RUS	SSP5	2050	0.121	-0.652	-0.256	0.175	0.078	0.089	0.158	0.434	0.329	0.139	0.656	0.139	0.123
RUS	SSP5	2100	0.109	-2.439	-0.284	0.091	0.075	0.085	0.145	0.352	0.281	0.05	0.585	0.05	0.115
ZAF	SSP1	2015	0.227	0.088	-0.068	0.146	0.093	0.096	0.284	1.098	0.701	0.192	0.535	0.192	0.324
ZAF	SSP1	2050	0.21	0.072	-0.067	0.19	0.09	0.093	0.259	0.811	0.57	0.133	0.512	0.133	0.292
ZAF	SSP1	2100	0.183	0.048	-0.064	0.152	0.085	0.088	0.223	0.569	0.433	0.019	0.473	0.019	0.248
ZAF	SSP2	2015	0.227	0.088	-0.068	0.146	0.093	0.096	0.284	1.098	0.701	0.192	0.535	0.192	0.324
ZAF	SSP2	2050	0.211	0.073	-0.068	0.147	0.091	0.093	0.261	0.828	0.578	0.138	0.514	0.138	0.295
ZAF	SSP2	2100	0.185	0.05	-0.065	0.136	0.086	0.088	0.225	0.582	0.441	0.028	0.476	0.028	0.251
ZAF	SSP3	2015	0.227	0.088	-0.068	0.146	0.093	0.096	0.284	1.098	0.701	0.192	0.535	0.192	0.324
ZAF	SSP3	2050	0.214	0.076	-0.068	0.131	0.091	0.094	0.265	0.866	0.598	0.148	0.518	0.148	0.3
ZAF	SSP3	2100	0.197	0.061	-0.066	0.183	0.088	0.09	0.242	0.68	0.499	0.085	0.495	0.085	0.271

Table C1: Projected income elasticities by country, scenario, and year (continued)

Country	Scenario	Year	wht	rce	crl, mze, str	sgr	swt, stm, alc	pls, nut, ocr, spc	vol	vgt	frt	cmt, rmt, omt	pmt	omt	mlk, dai, egg
ZAF	SSP4	2015	0.227	0.088	-0.068	0.146	0.093	0.096	0.284	1.098	0.701	0.192	0.535	0.192	0.324
ZAF	SSP4	2050	0.212	0.074	-0.068	0.154	0.091	0.093	0.261	0.831	0.58	0.139	0.514	0.139	0.295
ZAF	SSP4	2100	0.186	0.05	-0.065	0.146	0.086	0.088	0.227	0.588	0.445	0.033	0.478	0.033	0.252
ZAF	SSP5	2015	0.227	0.088	-0.068	0.146	0.093	0.096	0.284	1.098	0.701	0.192	0.535	0.192	0.324
ZAF	SSP5	2050	0.207	0.07	-0.067	0.214	0.09	0.092	0.255	0.779	0.553	0.123	0.508	0.123	0.287
ZAF	SSP5	2100	0.175	0.041	-0.063	0.132	0.084	0.086	0.214	0.522	0.403	-0.023	0.46	-0.023	0.237
TUR	SSP1	2015	0.124	-0.295	-0.304	0.222	0.129	0.011	0.129	0.245	0.214	0.582	0.958	0.582	0.117
TUR	SSP1	2050	0.116	-0.364	-0.336	0.07	0.123	0.011	0.123	0.223	0.197	0.378	0.8	0.378	0.111
TUR	SSP1	2100	0.112	-0.424	-0.359	0.082	0.119	0.011	0.119	0.212	0.189	0.287	0.738	0.287	0.108
TUR	SSP2	2015	0.124	-0.295	-0.304	0.222	0.129	0.011	0.129	0.245	0.214	0.582	0.958	0.582	0.117
TUR	SSP2	2050	0.117	-0.358	-0.334	0.068	0.123	0.011	0.123	0.224	0.198	0.389	0.808	0.389	0.112
TUR	SSP2	2100	0.112	-0.415	-0.356	0.083	0.12	0.011	0.12	0.213	0.19	0.298	0.746	0.298	0.109
TUR	SSP3	2015	0.124	-0.295	-0.304	0.222	0.129	0.011	0.129	0.245	0.214	0.582	0.958	0.582	0.117
TUR	SSP3	2050	0.117	-0.35	-0.33	0.068	0.124	0.011	0.124	0.226	0.2	0.407	0.821	0.407	0.112
TUR	SSP3	2100	0.113	-0.399	-0.35	0.117	0.12	0.011	0.12	0.216	0.192	0.32	0.76	0.32	0.109
TUR	SSP4	2015	0.124	-0.295	-0.304	0.222	0.129	0.011	0.129	0.245	0.214	0.582	0.958	0.582	0.117
TUR	SSP4	2050	0.116	-0.362	-0.335	0.198	0.123	0.011	0.123	0.223	0.197	0.382	0.803	0.382	0.111
TUR	SSP4	2100	0.112	-0.415	-0.356	0.083	0.12	0.011	0.12	0.213	0.19	0.299	0.746	0.299	0.109
TUR	SSP5	2015	0.124	-0.295	-0.304	0.222	0.129	0.011	0.129	0.245	0.214	0.582	0.958	0.582	0.117
TUR	SSP5	2050	0.115	-0.375	-0.341	0.095	0.122	0.011	0.122	0.22	0.195	0.357	0.785	0.357	0.111
TUR	SSP5	2100	0.108	-0.497	-0.383	0.088	0.116	0.011	0.116	0.203	0.182	0.218	0.696	0.218	0.106
USA	SSP1	2015	0.138	-0.106	-0.131	0.098	0.057	0.032	0.102	0.432	0.325	0.025	0.535	0.025	0.092
USA	SSP1	2050	0.134	-0.115	-0.132	0.19	0.056	0.031	0.1	0.396	0.305	-0.011	0.514	-0.011	0.09
USA	SSP1	2100	0.129	-0.124	-0.132	0.106	0.055	0.031	0.098	0.367	0.288	-0.046	0.494	-0.046	0.088
USA	SSP2	2015	0.138	-0.106	-0.131	0.098	0.057	0.032	0.102	0.432	0.325	0.025	0.535	0.025	0.092
USA	SSP2	2050	0.134	-0.113	-0.131	0.194	0.056	0.032	0.1	0.401	0.308	-0.005	0.518	-0.005	0.09
USA	SSP2	2100	0.131	-0.122	-0.132	0.112	0.056	0.031	0.098	0.375	0.292	-0.036	0.5	-0.036	0.089
USA	SSP3	2015	0.138	-0.106	-0.131	0.098	0.057	0.032	0.102	0.432	0.325	0.025	0.535	0.025	0.092
USA	SSP3	2050	0.135	-0.112	-0.131	0.2	0.056	0.032	0.101	0.408	0.312	0.002	0.522	0.002	0.09
USA	SSP3	2100	0.134	-0.114	-0.131	0.188	0.056	0.032	0.1	0.401	0.307	-0.006	0.517	-0.006	0.09
USA	SSP4	2015	0.138	-0.106	-0.131	0.098	0.057	0.032	0.102	0.432	0.325	0.025	0.535	0.025	0.092
USA	SSP4	2050	0.134	-0.115	-0.132	0.198	0.056	0.031	0.1	0.395	0.304	-0.012	0.514	-0.012	0.09
USA	SSP4	2100	0.13	-0.123	-0.132	0.107	0.056	0.031	0.098	0.371	0.29	-0.04	0.498	-0.04	0.088
USA	SSP5	2015	0.138	-0.106	-0.131	0.098	0.057	0.032	0.102	0.432	0.325	0.025	0.535	0.025	0.092
USA	SSP5	2050	0.133	-0.116	-0.132	0.131	0.056	0.031	0.099	0.391	0.302	-0.017	0.511	-0.017	0.09
USA	SSP5	2100	0.125	-0.134	-0.132	0.109	0.055	0.031	0.096	0.343	0.273	-0.082	0.475	-0.082	0.087
VNM	SSP1	2015	0.552	0.068	-0.521	0.287	0.287	0.014	0.669	0.455	0.119	0.183	0.974	0.183	0.72
VNM	SSP1	2050	0.432	0.06	-0.864	0.244	0.244	0.013	0.484	0.358	0.116	0.124	0.832	0.124	0.511
VNM	SSP1	2100	0.385	0.053	-2.52	0.226	0.226	0.013	0.426	0.323	0.123	0.092	0.772	0.092	0.448
VNM	SSP2	2015	0.552	0.068	-0.521	0.287	0.287	0.014	0.669	0.455	0.119	0.183	0.974	0.183	0.72
VNM	SSP2	2050	0.438	0.061	-0.814	0.246	0.246	0.013	0.493	0.363	0.116	0.128	0.84	0.128	0.521
VNM	SSP2	2100	0.389	0.055	-4.084	0.227	0.227	0.013	0.43	0.325	0.121	0.094	0.776	0.094	0.452
VNM	SSP3	2015	0.552	0.068	-0.521	0.287	0.287	0.014	0.669	0.455	0.119	0.183	0.974	0.183	0.72
VNM	SSP3	2050	0.453	0.062	-0.735	0.252	0.252	0.013	0.513	0.375	0.117	0.137	0.858	0.137	0.544
VNM	SSP3	2100	0.429	0.06	-0.888	0.243	0.243	0.013	0.48	0.355	0.116	0.122	0.828	0.122	0.507
VNM	SSP4	2015	0.552	0.068	-0.521	0.287	0.287	0.014	0.669	0.455	0.119	0.183	0.974	0.183	0.72
VNM	SSP4	2050	0.446	0.062	-0.77	0.249	0.249	0.013	0.503	0.369	0.116	0.132	0.849	0.132	0.532
VNM	SSP4	2100	0.398	0.058	-1.916	0.23	0.23	0.013	0.439	0.331	0.116	0.1	0.786	0.1	0.462
VNM	SSP5	2015	0.552	0.068	-0.521	0.287	0.287	0.014	0.669	0.455	0.119	0.183	0.974	0.183	0.72
VNM	SSP5	2050	0.423	0.06	-0.953	0.24	0.24	0.013	0.471	0.35	0.116	0.118	0.82	0.118	0.498
VNM	SSP5	2100	0.356	0.044	-2.813	0.217	0.217	0.013	0.398	0.306	0.138	0.072	0.739	0.072	0.418